Global exact controllability for Schrödinger equations with Lipschitz and logarithmic nonlinearities

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Abstract

In this paper, we study global exact controllability of semilinear Schrödinger equations by localized internal additive controls. For every globally Lipschitz nonlinearity in arbitrary dimensions, we prove that any two L^2 states can be connected in any prescribed positive time, without smallness assumptions on the states or the Lipschitz constant. The result holds on smooth bounded Euclidean domains under a local convex-phase condition, and on an interval with any nonempty open control region. It also extends to compact Riemannian manifolds under the corresponding phase condition. In one dimension, we further establish controllability for logarithmic nonlinearities of the form

$$f(z) = z\phi(|z|) + \mathfrak{g}(z),$$

where ϕ is real-valued, $\mathfrak{g}$ is globally Lipschitz, and $r|\phi'(r)| \le D(1 + |\log r|)^\gamma$ with $0 \le \gamma < 1/2$. For the endpoint case $\gamma = 1/2$, we achieve the result by assuming the leading coefficient is sufficiently small. These results resolve the global Lipschitz controllability question raised by Zuazua [29, Section 3] in the stated geometric settings and establish its logarithmic extension within the specified one-dimensional classes.

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1 Introduction

Our main goal is to establish global exact controllability of semilinear Schrödinger equations between arbitrary L^2 states in every prescribed positive time, using localized internal additive controls. Let $\Omega \subset \mathbb{R}^n$, $n \ge 1$, be a bounded, connected domain with C^∞ boundary, and let $\omega \subset \Omega$ be a nonempty open set. For $T > 0$, set

$$Q = (0,T) \times \Omega, \qquad Q_\omega = (0,T) \times \omega, \qquad \Sigma = (0,T) \times \partial\Omega.$$

We consider the controlled equation

$$\begin{cases} i\partial_t y = \Delta y + f(y) + \mathbf{1}_\omega h & \text{in } Q, \\ y = 0 & \text{on } \Sigma, \\ y(0) = y_0 & \text{in } \Omega. \end{cases} \tag{1.1}$$

The state y and control h are complex valued, and $\mathbf{1}_\omega h$ denotes the extension of h by zero to Ω. More precisely, for every prescribed $T > 0$ and every pair $y_0, y_1 \in L^2(\Omega;\mathbb{C})$, we seek a control $h \in L^2(Q_\omega;\mathbb{C})$ such that a solution of (1.1) satisfies $y(T) = y_1$. The time T is fixed independently of the endpoint norms. We treat globally Lipschitz nonlinearities in arbitrary dimension and nonlinearities with logarithmic growth in one dimension.

The questions raised in [29, Section 3] concern the passage from sufficiently small Lipschitz perturbations of the free equation to general globally Lipschitz nonlinearities, and its extension to logarithmically superlinear growth in arbitrary positive time; see also the discussion in [22, Section 6.2.2]. For (1.1), these questions ask whether, under suitable geometric assumptions on (Ω, ω), arbitrary $L^2(\Omega;\mathbb{C})$ states can be joined in every prescribed positive time in the following cases:

(a) $f : \mathbb{C} \to \mathbb{C}$ is globally Lipschitz, with no smallness restriction imposed on its Lipschitz constant;

(b) f exhibits logarithmic superlinear growth, including the example $f(z) = \lambda z \log|z|^2$ for $z \neq 0$, with $f(0) = 0$ and $\lambda \in \mathbb{R}$, in the sense of a suitable low-regularity solution concept.

The difficulty in addressing the above two questions lies in closing the nonlinear control argument for arbitrary Lipschitz constants and endpoint norms. Recall that a contraction argument based on fixed unweighted estimates necessitates a small Lipschitz constant. An alternative is to freeze the nonlinearity, which yields bounded time-dependent potentials; this approach requires uniform observability together with a nonlinear fixed-point construction at the L^2 regularity level [29, (3.7)–(3.8)]. Genuinely different from parabolic equations, the unitary Schrödinger group provides no Sobolev regularization for L^2 solutions, rendering the standard compactness argument inapplicable. In Section 1.1, we shall compare this approach and existing controllability results with the weighted contraction argument developed herein.

The main contribution of this work is to provide an affirmative answer to the above two questions under a local convex-phase condition. More precisely, our first main result, Theorem 1.4, resolves question **(a)** on domains satisfying a local convex-phase condition near the uncontrolled region, and on a spatial interval with any nonempty open control region. This result eliminates the smallness restriction inherent in elementary perturbation arguments, allowing for arbitrary L^2 terminal states and all time horizons $T > 0$. In particular, no differentiability, gauge invariance, or conservation law assumptions are imposed on the nonlinearity f. Theorems 1.8 and 1.9 answer question **(b)** on a spatial interval for the logarithmic nonlinearity classes specified hereafter. In particular, these results cover $f(z) = \lambda z \log|z|^2$ for all $\lambda \in \mathbb{R}$, as well as arbitrary globally Lipschitz perturbations. Analogous to the Lipschitz setting, the control time is independent of the terminal-state norms.

1.1 Comparison with existing works. We shall compare our results with closely related works on control and observability for Schrödinger equations, and the comparison mainly concerns the nonlinearities, endpoint spaces, geometric assumptions and dependence of the control time on the data.

For the cubic Schrödinger equation on a periodic interval, Illner et al. [15] established local exact controllability in H^1. In the case of a bounded interval with periodic, Dirichlet or Neumann

boundary conditions, Rosier and Zhang [25] obtained local exact controllability in L^2. They also treated polynomial nonlinearities on rectangles in compatible Sobolev spaces, where the required regularity depends on the dimension, degree and boundary conditions [26]. In these results, the positive control time is prescribed first, and the endpoints must lie in a sufficiently small neighborhood of zero. Moreover, the rectangular theory allows any nonempty open internal control region, including regions that do not satisfy the geometric control condition (GCC). For the globally Lipschitz class considered here, Theorem 1.4 establishes exact controllability between arbitrary L^2 states without restricting their norms or the Lipschitz constant.

Global exact controllability has also been obtained by combining stabilization with local control. On an interval, Laurent [20] proved global exact controllability of the cubic equation with either sign of the nonlinearity in L^2, using any nonempty open control region. More precisely, for each $R > 0$, a common control time is obtained for all endpoints with norm at most R. For the defocusing equation on compact surfaces, Dehman et al. [11] established the corresponding H^1 theory under GCC and an additional unique-continuation assumption. In dimension three, Laurent [21] proved global exact controllability for the defocusing cubic equation in H^1 under these assumptions and a spectral bilinear estimate; the result also covers cuboids by reflection.

Concerning the dependence of the control time on the size of the data, Le Balc'h and Martin [23] obtained uniform exponential energy stabilization for the defocusing cubic equation on tori of dimension at most three, provided that the control region contains a strip in each coordinate direction. Their estimate gives a logarithmic upper bound for the null-control time in terms of the initial energy. Under GCC, Loyola [24] removed the additional nonlinear unique-continuation assumption for gauge-invariant defocusing polynomial models on compact surfaces and defocusing cubic models on the specified three-dimensional manifolds, obtaining semiglobal H^1 exact controllability. Thus, for each bounded set of endpoints, these results provide a common sufficiently large control time. In Theorem 1.4, the time $T > 0$ is prescribed independently of the endpoint norms for every globally Lipschitz map $f : \mathbb{C} \to \mathbb{C}$, including $f(0) \neq 0$, without requiring differentiability or a conservation law.

In addition to the nonlinear assumptions and control time, the geometric conditions in these results differ from those used here. The above manifold results rely on GCC, which concerns the geodesic flow, whereas Assumption 1.2 requires a smooth real phase with a nonvanishing gradient and a positive definite Hessian near $K_\omega = \overline{\Omega \setminus \omega}^{\Omega}$, with $\partial_\nu \psi = 0$ on the relevant boundary neighborhood. Thus the phase is required to be convex and noncritical only near the uncontrolled set. Under the corresponding covariant condition, Proposition 1.11 gives exact control between arbitrary L^2 states on compact Riemannian manifolds in every prescribed positive time. On an interval, Theorem 1.4(ii) allows every nonempty open control region without a phase assumption.

To pass from linear to nonlinear control by freezing the nonlinearity, one also needs to track the dependence of observability estimates on the potential. Bourgain et al. [6] proved observability and additive controllability on two-dimensional tori for a fixed real, time-independent L^2 potential, from any nonempty open set in every positive time. On $\mathbb{R}$, Su et al. [27] obtained quantitative observability for real, time-independent bounded potentials on thick sets, with a bound of the form Ce^{C/T^2}, where the constants are controlled by the potential norm and thickness parameters. These estimates concern fixed spatial potentials. After separating $f(0)$, however, freezing a globally Lipschitz nonlinearity gives a family of bounded time-dependent potentials that need not be real. An estimate for a fixed potential therefore does not by itself give the uniform observability required for this family.

For this approach, Zuazua [29, (3.7)–(3.8)] identified two requirements: uniform observability and compactness of the nonlinear fixed-point map. The compactness issue arises because the unitary Schrödinger group provides no Sobolev regularization of L^2 data, and hence

the compactness argument based on parabolic smoothing does not apply. To address the nonlinear step, Zuazua also suggested the global inverse function approach used by Lasiecka and Triggiani [17]. For the observability step, Lasiecka et al. [18, 19] established Carleman estimates for nonconservative Schrödinger equations at the H^1 and L^2 levels, and Triggiani and Xu [28] developed an H^1 theory on Riemannian manifolds. In the present work, a parameter-uniform zeroth-order Carleman estimate allows us to decrease the source-to-state gain by choosing the parameter sufficiently large. This makes the nonlinear correction contractive in weighted L^2 spaces for every finite Lipschitz constant and yields the control cost $Ce^{C(T^{-1}+L^{2/3})}$ in (1.5), without requiring compactness of the fixed-point map.

For logarithmic nonlinearities, Beauchard et al. [3] studied $f(z) = \lambda z \log |z|^2$ for arbitrary real λ on tori and Euclidean spaces. Using finitely many real bilinear potentials, they established global L^2 approximate controllability between arbitrary states on the unit sphere, up to a constant phase, in arbitrarily short times. Theorems 1.8 and 1.9 concern localized additive controls on an interval, where any nonempty open control region is allowed and arbitrary L^2 endpoints can be joined exactly in every prescribed time $T > 0$. Moreover, these theorems apply to $f(z) = z\phi(|z|) + \mathfrak{g}(z)$, where $\phi \in W^{1,\infty}_{\mathrm{loc}}((0,\infty);\mathbb{R})$ and $\mathfrak{g}$ is any globally Lipschitz map, with the radial term extended by zero at the origin. Under $r|\phi'(r)| \le D(1+|\log r|)^\gamma$, every finite D is allowed when $0 \le \gamma < 1/2$. At the critical exponent, the condition $r|\phi'(r)| \le D_0 + D_1\sqrt{1+|\log r|}$ requires only D_1 to be sufficiently small, while D_0 and the Lipschitz perturbation remain unrestricted. In particular, the canonical logarithmic term is covered for every real coefficient, and the controlled solutions belong to $C([0,T];L^2) \cap L^4(Q)$ with $f(y) \in L^2(Q)$. The estimates (1.8) and (1.12) quantify the dependence of the control cost on time and endpoint size; for the canonical term, (1.10) gives a linear dependence on the endpoint norms.

1.2 Main results. Let us first state the assumptions on the nonlinearity and the control geometry for (1.1).

Assumption 1.1. The function $f : \mathbb{C} \to \mathbb{C}$ is globally Lipschitz: for some finite $L > 0$,

$$|f(z) - f(w)| \le L|z - w| \qquad \text{for all } z, w \in \mathbb{C}. \tag{1.2}$$

Assumption 1.2. Let $K_\omega = \overline{\Omega \setminus \omega}^{\,\overline{\Omega}}$. There exist a relatively open neighborhood $\mathcal{O}$ of K_ω in $\overline{\Omega}$, a real-valued function ψ smooth on a neighborhood of $\overline{\Omega}$, and constants $c_0, c_1 > 0$ such that

$$|\nabla\psi| \ge c_0, \quad D^2\psi \ge c_1 I \quad \text{on } \mathcal{O}, \qquad \partial_\nu\psi = 0 \quad \text{on } \mathcal{O} \cap \partial\Omega. \tag{1.3}$$

Here ν is the outward unit normal to $\partial\Omega$, and the Hessian inequality is understood in the sense of quadratic forms.

For the Euclidean Dirichlet problem, set

$$A = -\Delta_D \quad \text{on } H := L^2(\Omega;\mathbb{C}), \qquad D(A) = H^2(\Omega;\mathbb{C}) \cap H^1_0(\Omega;\mathbb{C}), \qquad U(t) = e^{itA}.$$

The boundary condition is encoded in A, and the Schrödinger operator is given by $P = i\partial_t - \Delta = i\partial_t + A$. A mild solution of (1.1) is a function $y \in C([0,T];H)$ satisfying

$$y(t) = U(t)y_0 - i\int_0^t U(t-r)(f(y(r)) + \mathbf{1}_\omega h(r))\,\mathrm{d}r. \tag{1.4}$$

We use the same symbol f for its Nemytskii map on H, defined by $f(v)(x) = f(v(x))$. Since Ω has finite measure, (1.2) implies that this map is globally Lipschitz on H and that

$$\|f(v)\|_H \le L\|v\|_H + |\Omega|^{1/2}|f(0)|.$$

It follows that, for every $y_0 \in H$ and $h \in L^2(Q_\omega;\mathbb{C})$, equation (1.1) has a unique mild solution $y \in C([0,T];H)$.

Definition 1.3. The system (1.1) is said to be *globally exactly controllable* in H if, for any $T > 0$ and all endpoints $y_0, y_1 \in H$, there exists a control $h \in L^2(Q_\omega;\mathbb{C})$ such that the corresponding mild solution y of (1.1) satisfies $y(T) = y_1$.

We now state the first main result for globally Lipschitz nonlinearities.

Theorem 1.4. *Let f satisfy Assumption 1.1, and assume that one of the following holds:*

(i) *Ω and ω satisfy Assumption 1.2;*

(ii) *$\Omega = (0,\ell)$ for some $\ell > 0$, and $\omega \subset (0,\ell)$ is any nonempty open set.*

Then (1.1) is globally exactly controllable in the sense of Definition 1.3. Moreover, the control may be chosen so that

$$\|h\|_{L^2(Q_\omega)} + \|y\|_{C([0,T];H)} \le Ce^{C(T^{-1}+L^{2/3})}(\|y_0\|_H + \|y_1\|_H + T|\Omega|^{1/2}|f(0)|). \tag{1.5}$$

The constant C depends only on the fixed spatial geometry, the control region, and the phase data. It is independent of T, L, f and the endpoints.

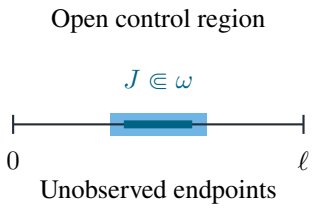
Example 1.5. Theorem 1.4 applies to the following control geometries; see Figure 1.

(i) $\Omega = (0,\ell)$, $\ell > 0$, and ω is any nonempty open subset of Ω.

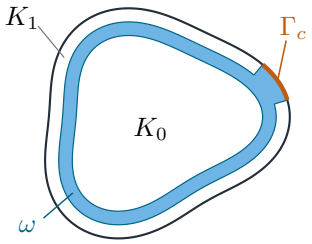
(ii) Let $\Omega \subset \mathbb{R}^2$ be a bounded simply connected domain with C^∞ boundary, and let $\Gamma_c \subset \partial\Omega$ be a nonempty open arc with $\overline{\Gamma_c} \neq \partial\Omega$. Set $d(x) = \operatorname{dist}(x,\partial\Omega)$, and let π be the nearest-point projection in a tubular neighborhood of $\partial\Omega$. For sufficiently small $\delta > 0$, with 2δ below the tubular radius, define

$$\omega = \{x \in \Omega : \delta < d(x) < 2\delta\} \cup \{x \in \Omega : 0 < d(x) < 2\delta,\ \pi(x) \in \Gamma_c\}. \tag{1.6}$$

This region satisfies Assumption 1.2; the verification is given in Appendix A.



(a) An interval



(b) A planar domain

Figure 1: Schematic control geometries in (a) one dimension and (b) two dimensions. The control regions are shown in blue. The sets K_0, K_1 and the planar phase construction are given in Appendix A.

On an interval, we also consider nonlinearities that need not be locally Lipschitz at zero and may grow faster than linearly. We make the following assumption.

Assumption 1.6. There exist constants $0 \le \gamma < 1/2$, $0 \le D < \infty$, a real-valued function $\phi \in W^{1,\infty}_{\mathrm{loc}}((0,\infty))$, and a globally Lipschitz mapping $\mathfrak{g} : \mathbb{C} \to \mathbb{C}$ such that

$$f(z) = z\phi(|z|) + \mathfrak{g}(z) \quad (z \neq 0), \qquad f(0) = \mathfrak{g}(0),$$

and

$$r|\phi'(r)| \le D(1 + |\log r|)^{\gamma} \quad \text{for almost every } r > 0. \tag{1.7}$$

We use the locally Lipschitz representative of ϕ. The value $\mathfrak{g}(0) \in \mathbb{C}$ is arbitrary, and no sign condition is imposed on the imaginary part of $\mathfrak{g}(z)\bar{z}$.

Remark 1.7. (1) The growth assumption gives

$$|\phi(r)| \le |\phi(1)| + CD(1 + |\log r|)^{1+\gamma},$$

so the radial term extends continuously by zero at the origin. When $\gamma = 0$, the map $q \mapsto \phi(e^q)$ is globally Lipschitz. The reality assumption concerns only the radial coefficient; the full nonlinearity may be complex-valued.

(2) Assumption 1.6 applies to the canonical logarithmic nonlinearity as well as to

$$\lambda z \log(1 + |z|^2), \quad \lambda z(\log(2 + |z|^2))^{\alpha}, \quad \lambda \in \mathbb{R},\ 1 < \alpha < 3/2,$$

and to their arbitrary globally Lipschitz perturbations.

The second main result concerns this logarithmic class.

Theorem 1.8. *Let $\Omega = (0, \ell)$ with $\ell > 0$, and let $\omega \subset \Omega$ be nonempty and open. Suppose that Assumption 1.6 holds. For every $T > 0$ and $y_0, y_1 \in H$, there exists a control $h \in L^2(Q_\omega; \mathbb{C})$ such that (1.1) admits a Dirichlet mild solution in the sense of (1.4), satisfying*

$$y \in C([0,T]; H) \cap L^4(Q), \qquad f(y) \in L^2(Q),$$
$$y(0) = y_0, \qquad y(T) = y_1.$$

For this control and initial state, the solution is unique within $C([0,T]; H) \cap L^4(Q)$. When $\gamma = 0$, uniqueness also holds among mild solutions in $C([0,T]; H)$ with $f(y) \in L^1(0,T; H)$.

Set $d = \|y_0\|_H + \|y_1\|_H$. The following estimates hold:

(i) *Let $q = 2(1+\gamma)/3 < 1$. The control may be selected such that*

$$\|h\|_{L^2(Q_\omega)} + \|y\|_{C([0,T];H)} \le Ce^{C/T}\left((1+d)\exp(C[\log(2+d)]^q) + \sqrt{T}\right). \tag{1.8}$$

Consequently, for any $\varepsilon > 0$,

$$\|h\|_{L^2(Q_\omega)} + \|y\|_{C([0,T];H)} \le C_\varepsilon e^{C/T}((1+d)^{1+\varepsilon} + \sqrt{T}). \tag{1.9}$$

Here C depends only on Ω, ω, γ, D, $|\phi(1)|$, $\mathrm{Lip}(\mathfrak{g})$ and $|\mathfrak{g}(0)|$; C_ε additionally depends on ε. In particular, C is independent of ε, and both constants are independent of T, y_0, y_1.

(ii) *If $f(z) = \lambda z \log|z|^2$ for $z \neq 0$, with $f(0) = 0$ and $\lambda \in \mathbb{R}$, then the control can instead be chosen so that*

$$\|h\|_{L^2(Q_\omega)} + \|y\|_{C([0,T];H)} \le C_\lambda e^{C_\lambda/T} d, \tag{1.10}$$

where C_λ depends only on Ω, ω and λ.

At the exponent $\gamma = 1/2$, the argument requires smallness only of the leading coefficient.

Theorem 1.9. *Let $\Omega = (0, \ell)$, $\ell > 0$, and let $\omega \subset \Omega$ be nonempty and open. There are constants $D_*, c_* > 0$, depending only on Ω and ω, with the following property. Suppose that*

$$f(z) = z\phi(|z|) + \mathfrak{g}(z) \quad (z \neq 0), \qquad f(0) = \mathfrak{g}(0),$$

where $\phi \in W^{1,\infty}_{\mathrm{loc}}((0,\infty);\mathbb{R})$ is taken in its locally Lipschitz representative and $\mathfrak{g} : \mathbb{C} \to \mathbb{C}$ is globally Lipschitz. Assume that

$$r|\phi'(r)| \leq D_0 + D_1\sqrt{1 + |\log r|} \quad \text{for a.e. } r > 0, \qquad D_0 \geq 0, \quad 0 \leq D_1 \leq D_*. \tag{1.11}$$

For every $T > 0$ and $y_0, y_1 \in H$, there is a control $h \in L^2(Q_\omega;\mathbb{C})$ such that (1.1) admits a Dirichlet mild solution in the sense of (1.4), with

$$y \in C([0,T];H) \cap L^4(Q), \qquad f(y) \in L^2(Q),$$
$$y(0) = y_0, \qquad y(T) = y_1.$$

For this control and initial state, the solution is unique in $C([0,T];H) \cap L^4(Q)$. Writing $d = \|y_0\|_H + \|y_1\|_H$, the control can be chosen, for every $\varepsilon > 0$, so that

$$\|h\|_{L^2(Q_\omega)} + \|y\|_{C([0,T];H)} \leq C_\varepsilon e^{C/T}\left((1+d)^{1+c_*D_1^{2/3}+\varepsilon} + \sqrt{T}\right). \tag{1.12}$$

Here C depends only on Ω, ω, D_0, D_1, $|\phi(1)|$, $\mathrm{Lip}(\mathfrak{g})$ and $|\mathfrak{g}(0)|$; C_ε may also depend on ε. In particular, C is independent of ε, and both constants are independent of T, y_0, y_1. No smallness is required of D_0 or $\mathfrak{g}$.

Remark 1.10. Under the notation and regularity assumptions of Theorem 1.9, its conclusions remain valid if (1.11) is replaced by

$$\max\left\{\operatorname*{ess\,limsup}_{r\downarrow 0}\frac{r|\phi'(r)|}{\sqrt{1+|\log r|}}, \operatorname*{ess\,limsup}_{r\to\infty}\frac{r|\phi'(r)|}{\sqrt{1+|\log r|}}\right\} < D_*.$$

Indeed, choose $D_1 < D_*$ above both limits; local Lipschitz regularity allows the contribution from the remaining compact amplitude range to be absorbed into a finite D_0 in (1.11).

1.3 Extension to Riemannian manifolds. The weighted control argument extends to compact Riemannian manifolds under a covariant version of the local phase condition. The result again allows arbitrary L^2 endpoints in every prescribed positive time, without restricting the Lipschitz constant. Riemannian Carleman estimates for Schrödinger equations were developed by Triggiani and Xu [28] at the H^1 level. In our setting, the control acts in the interior, and the phase condition uses the covariant Hessian and tangency on the unobserved boundary.

Proposition 1.11. *Let (M, g) be a smooth compact connected Riemannian manifold, either with smooth boundary or without boundary, and let $\omega \subset M^\circ$ be nonempty and open, where M° denotes the interior. Suppose that there are a relatively open neighborhood $\mathcal{O}$ of $K_\omega = \overline{M^\circ \setminus \omega}^{M}$, a real $\psi \in C^\infty(M)$, and $c_0, c_1 > 0$ such that*

$$|\nabla_g \psi|_g \geq c_0, \qquad \mathrm{Hess}_g\,\psi \geq c_1 g \quad \text{on } \mathcal{O},$$
$$\partial_{\nu_g}\psi = 0 \quad \text{on } \mathcal{O} \cap \partial M. \tag{1.13}$$

The last condition is omitted if $\partial M = \varnothing$. If f satisfies Assumption 1.1, then for every $T > 0$ and $y_0, y_1 \in L^2(M;\mathbb{C})$, the equation

$$i\partial_t y = \Delta_g y + f(y) + \mathbf{1}_\omega h$$

has a control $h \in L^2((0,T)\times\omega;\mathbb{C})$ whose unique mild solution satisfies $y \in C([0,T];L^2(M;\mathbb{C}))$, $y(0) = y_0$ and $y(T) = y_1$. Homogeneous Dirichlet conditions are imposed when $\partial M \neq \varnothing$. Moreover,

$$\begin{aligned}\|h\|_{L^2((0,T)\times\omega)} + \|y\|_{C([0,T];L^2(M))} \\ \leq Ce^{C(T^{-1}+L^{2/3})}(\|y_0\|_{L^2(M)} + \|y_1\|_{L^2(M)} + T\,\mathrm{Vol}_g(M)^{1/2}|f(0)|).\end{aligned} \tag{1.14}$$

All spatial norms use the Riemannian volume measure. The constant C depends only on the fixed manifold, metric, control region and phase data; it is independent of T, L, f, y_0, y_1.

The proof is presented in Section 2.5. The local phase condition can be illustrated by an interior geodesic ball $B_g(p,R)$, chosen sufficiently small that $r(x)^2 = d_g(p,x)^2$ is smooth and $\mathrm{Hess}_g(r^2) \geq g$ on the ball. If K_ω lies in $\{a \leq r \leq b\}$ with $0 < a < b < R$, a smooth extension of r^2 near K_ω satisfies (1.13).

Corollary 1.12. *Let (M,g) be a smooth, compact, connected one-dimensional Riemannian manifold, possibly with boundary, and let $\omega \subset M^\circ$ be nonempty and open. The conclusions of Theorems 1.4, 1.8, and 1.9 hold for*

$$i\partial_t y - \Delta_g y = f(y) + \mathbf{1}_\omega h, \qquad y(0) = y_0, \quad y(T) = y_1,$$

with homogeneous Dirichlet conditions when $\partial M \neq \varnothing$. Here $H = L^2(M;\mathbb{C})$, all spatial integrals use the Riemannian volume, and the respective nonlinear assumptions are unchanged. The dependence of the constants on the interval and control region is replaced by dependence on (M,g) and ω; all other stated dependencies remain unchanged.

1.4 Main idea of the proof. The proof relies on a linear control operator whose source-to-state gain decreases as the Carleman parameter increases. For each prescribed $T > 0$, this gain can be made sufficiently small to obtain a contraction for any finite Lipschitz constant and arbitrary endpoint data. The logarithmic extension then follows by combining the resulting control-cost estimate with quantitative bounds on the truncation error. We describe the construction in three steps.

1. A zeroth-order Carleman estimate with a free parameter. We use the parameter-dependent weighted contraction strategy developed by Hernández-Santamaría et al. [14] for semilinear stochastic parabolic equations, where their Carleman construction adapts the time weights of Badra et al. [2] for nonhomogeneous incompressible Navier–Stokes equations. To the best of our knowledge, this strategy appears to be new for nonlinear Schrödinger equations. For Schrödinger equations, Lasiecka et al. [18, 19] developed Carleman estimates for observability. To apply the weighted contraction argument to the present exact controllability problem, we use weights singular at both time endpoints and establish a zeroth-order state estimate with a zeroth-order observation. The nonlinear argument requires both zeroth-order norms; a gradient estimate alone does not provide them.

To this end, we lower the order of a strong Carleman estimate by applying a Bessel regularizer with parameter $\kappa = s/[t(T-t)]$. The positive term has size κ^3, while the squared commutator errors are bounded by $C\kappa^2$. Under the local phase condition, the transport field is tangent to the unobserved boundary. The boundary flux therefore vanishes, and the Dirichlet resolvent commutator can be used. The interval case is treated separately by reflecting only the linear test functions. By tracking the constants in these estimates, we obtain the threshold $s \geq aT + bT^2$.

2. Global exact controllability from the parameter gain. Using the Carleman estimate and weighted Hilbert-space duality, we construct a linear solution operator with zero initial and final

states. Its norm from the weighted source space to the weighted state space is uniform in s. For fixed T, the embedding of the state space into the source space has norm $O(s^{-3/2})$, which provides the small factor needed for the nonlinear correction.

To accommodate arbitrary endpoints, we first join a forward uncontrolled trajectory starting from y_0 to a backward uncontrolled trajectory ending at y_1. The resulting residual is supported away from the time endpoints and hence belongs to the weighted source space, including when $f(0) \neq 0$. The nonlinear correction is then a contraction provided that $CLs^{-3/2} < 1$, with no restriction on the endpoint norms. By choosing s quantitatively and subdividing the time interval into finitely many pieces, we obtain the control-cost factor $e^{C(T^{-1}+L^{2/3})}$ without using a conservation law. The same construction gives a uniform control estimate for the shifted nonlinearity $f(u+z) - f(u)$ along a prescribed reference function u. We use this estimate to control the shifted logarithmic truncations in the next step.

3. The logarithmic extension by Lipschitz truncation. We first clamp $\log|z|$ to $[-K, K]$ to obtain a globally Lipschitz nonlinearity f_K with Lipschitz constant $O(K^{1+\gamma})$. The preceding control estimate gives a cost with exponent $C(\tau^{-1} + K^{2(1+\gamma)/3})$, while the truncation error satisfies

$$|f(z) - f_K(z)| \leq C(1+K)^{\gamma} e^{-K}(1+|z|^2).$$

We compare the original and truncated equations through a common truncation, so the comparison introduces only the factor $e^{C\tau K^{\gamma}}$. The remaining tail is controlled by the one-dimensional space–time L^4 estimate. Since $2(1+\gamma)/3 < 1$ when $\gamma < 1/2$, the truncation error decays faster than the control cost grows.

These estimates allow us to construct controls on shrinking time intervals whose squared norms are summable. The resulting controls drive the true solution toward an uncontrolled reference trajectory ending at the desired target. By tracking the first truncation level, we obtain a control cost with nearly linear dependence on the data.

At the endpoint $\gamma = 1/2$, the leading exponent in the control cost is linear in K, and the argument closes when its coefficient is sufficiently small. By keeping the lower-order term separate, we obtain the two-coefficient critical theorem. Separating a compact interval of amplitudes from the two tails then yields the asymptotic condition. For the canonical logarithmic nonlinearity, amplitude scaling gives the stronger control-cost bound.

1.5 Organization of the paper. The paper is organized as follows. Section 2 establishes the parameter-dependent Carleman estimates and their Riemannian extension. Section 3 constructs the weighted linear control operator and proves global exact controllability for globally Lipschitz nonlinearities, together with the quantitative control-cost estimates. Section 4 proves the logarithmic results in the subcritical and critical regimes. The planar phase condition for the control geometry in Example 1.5 is verified in Appendix A.

2 Parameter-dependent Carleman estimates

2.1 Weights and regularization. All complex scalar products are linear in the first variable. Unless otherwise specified, the spatial L^2 norms in full-space estimates are taken over $\mathbb{R}^n$. The constants may depend on the fixed domain, control region, cutoffs, spatial phase, and T. They are independent of s and $\kappa \geq 1$. Uniformity for all $T > 0$ is established in Proposition 2.12.

For the full-space estimates below and the collar estimate in Section 2.2, we choose the following quadratic phase independently of the local phase in (1.3). Choose a bounded open set U with $\overline{\Omega} \subset U$, a point $x_0 \notin \overline{U}$, and a constant M such that

$$\psi(x) = |x - x_0|^2 - M < -1 \quad (x \in \overline{U}). \tag{2.1}$$

There are constants $\alpha, \beta, c_* > 0$ such that

$$1 < \alpha \le -\psi \le \beta, \qquad |\nabla \psi| \ge c_*, \qquad D^2\psi = 2I \quad \text{on } \overline{U}.$$

Define

$$a(t) = \frac{1}{t(T-t)}, \qquad \kappa(t) = sa(t), \qquad \theta(t,x) = e^{\kappa(t)\psi(x)}.$$

The dependence of θ on s is left implicit.

Lemma 2.1. *For $s > 0$ and $0 < t < T$,*

$$\kappa \ge \frac{4s}{T^2}, \qquad |\kappa'| \le \frac{T}{s}\kappa^2, \qquad 0 < \kappa'' \le \frac{2T^2}{s^2}\kappa^3. \tag{2.2}$$

In particular,

$$|\kappa''| + \frac{|\kappa'|^2}{\kappa} \le \frac{3T^2}{s^2}\kappa^3.$$

For each fixed $s > 0$, the product of any polynomial in κ with θ tends to zero at both temporal endpoints, uniformly on $\overline{U}$.

Proof. Put $d = t(T-t)$. Direct differentiation gives

$$\kappa' = -s(T-2t)d^{-2}, \qquad \kappa'' = 2sd^{-2} + 2s(T-2t)^2 d^{-3}.$$

It follows from the identity $d + (T-2t)^2 = T^2 - 3t(T-t) \le T^2$ that (2.2) holds. Finally, $\theta \le e^{-\alpha\kappa}$ shows that exponential decay dominates every polynomial in κ at the temporal endpoints. □

Lemma 2.2. *For $\kappa \ge 1$, let*

$$S_\kappa = \kappa(\kappa^2 - \Delta_{\mathbb{R}^n})^{-1/2}.$$

Then

$$\|S_\kappa\|_{2\to 2} \le 1, \qquad \|S_\kappa \operatorname{div}\|_{L^2(\mathbb{R}^n;\mathbb{C}^n)\to L^2(\mathbb{R}^n)} \le \kappa, \qquad \|\partial_\kappa S_\kappa\|_{2\to 2} \le C\kappa^{-1}. \tag{2.3}$$

For $b \in C_b^\infty(\mathbb{R}^n)$,

$$\|[b, S_\kappa]\|_{2\to 2} \le C\kappa^{-1}\|\nabla b\|_\infty, \qquad \|[b, S_\kappa]\partial_j\|_{2\to 2} \le C\|\nabla b\|_\infty. \tag{2.4}$$

Moreover, for every $v \in L^2(\mathbb{R}^n)$,

$$\kappa\|\nabla S_\kappa v\|_2^2 + \kappa^3\|S_\kappa v\|_2^2 = \kappa^3\|v\|_2^2. \tag{2.5}$$

The commutators are initially defined on $C_c^\infty(\mathbb{R}^n)$ and extended by continuity. The constants depend only on the dimension, apart from the displayed coefficient norm.

Proof. The multiplier of S_κ is

$$m_\kappa(\xi) = \frac{\kappa}{(\kappa^2 + |\xi|^2)^{1/2}}.$$

The bounds in (2.3) and the identity (2.5) follow from Plancherel's theorem. For the parameter derivative, we use

$$\partial_\kappa m_\kappa(\xi) = \frac{|\xi|^2}{(\kappa^2 + |\xi|^2)^{3/2}}.$$

To prove (2.4), we use the positive convolution kernel

$$K_\kappa(z) = \frac{\kappa}{\sqrt{\pi}} \int_0^\infty r^{-1/2} e^{-\kappa^2 r} (4\pi r)^{-n/2} e^{-|z|^2/(4r)} \, \mathrm{d}r. \tag{2.6}$$

The heat-kernel moment estimates give

$$\int_{\mathbb{R}^n} K_\kappa \, \mathrm{d}z = 1, \qquad \int_{\mathbb{R}^n} |z| K_\kappa(z) \, \mathrm{d}z \le C\kappa^{-1}, \qquad \int_{\mathbb{R}^n} |z| |\nabla K_\kappa(z)| \, \mathrm{d}z \le C. \tag{2.7}$$

Indeed, the first moment of the heat kernel is $C_n\sqrt{r}$, whereas the integral of $|z|$ times the magnitude of its gradient is uniform in r. Substituting these bounds into (2.6) gives (2.7).

The kernel of $[b, S_\kappa]$ is

$$(b(x) - b(y))K_\kappa(x - y).$$

The first commutator bound follows from $|b(x) - b(y)| \le \|\nabla b\|_\infty |x - y|$ and Young's inequality. To obtain the second bound, we take $v \in C_c^\infty(\mathbb{R}^n)$ and integrate by parts in y_j. The resulting kernel is the sum of

$$(\partial_j b)(y) K_\kappa(x - y), \qquad (b(x) - b(y))(\partial_j K_\kappa)(x - y).$$

The moment bounds in (2.7) control both kernels on L^2. We justify the integration by parts by first truncating the heat-kernel integral away from $r = 0$ and then removing the truncation using the same bounds. Density gives the required extensions. □

Lemma 2.3. *Let $K \Subset U$, and let $\zeta \in C_c^\infty(U)$ equal one on a fixed neighborhood of K. For every integer $N \ge 1$, there is $C_N > 0$, depending only on n, N, K, and ζ, such that*

$$\|(1 - \zeta) S_\kappa v\|_{H^1(\mathbb{R}^n)} \le C_N \kappa^{-N} \|v\|_2 \quad (v \in L^2(\mathbb{R}^n), \ \operatorname{supp} v \subset K, \ \kappa \ge 1). \tag{2.8}$$

If $\eta \in C_c^\infty(\mathbb{R}^n)$ has support at a positive distance from K, the same bound holds for $\|\eta S_\kappa v\|_2 + \|\eta \nabla S_\kappa v\|_2$, with a constant that may also depend on η.

Proof. For $d > 0$ and $|\gamma| \le 1$, the kernel representation (2.6) implies

$$\int_{\{z \in \mathbb{R}^n : |z| \ge d\}} |\partial_z^\gamma K_\kappa(z)| \, \mathrm{d}z \le C_{N,d} \kappa^{-N}.$$

The heat-kernel tail contributes $e^{-d^2/(8r)}$, and $\kappa^2 r + d^2/(8r)$ is bounded below by a positive multiple of κd. The resulting exponential decay absorbs the polynomial factors from the remaining integrals. Since $x \notin \{\zeta = 1\}$ and $y \in K$ are separated by a fixed positive distance, Young's inequality gives (2.8), both before and after one spatial derivative. □

2.2 A boundary collar. We first consider a control region containing a full boundary collar:

$$\{x \in \Omega : \operatorname{dist}(x, \partial\Omega) < \delta\} \subset \omega \quad \text{for some } \delta > 0. \tag{2.9}$$

This geometry satisfies Assumption 1.2. Indeed, K_ω has a neighborhood compactly contained in Ω, on which $\psi(x) = |x - x_*|^2$, with $x_* \notin \overline{\Omega}$, is strictly convex and has no critical point. The boundary condition on the phase is then vacuous. We retain the weights of Section 2.1 and set $P = i\partial_t - \Delta$. All integrals below are taken with respect to Lebesgue measure. Constants may depend on the fixed sets, cutoffs, weight, and T. They do not depend on s once its stated lower bound is imposed.

Proposition 2.4. *Assume (2.9). There are $s_0 > 0$ and $C > 0$ such that, for every $s \geq s_0$ and every $p \in C^\infty([0,T] \times \overline{\Omega}; \mathbb{C})$,*

$$\int_Q \kappa^3\theta^2|p|^2 \leq C\left(\int_Q \theta^2|Pp|^2 + \int_{Q_\omega} \kappa^3\theta^2|p|^2\right). \tag{2.10}$$

The test function p is unrestricted at $t = 0, T$ and on the spatial boundary.

We first prove an estimate for functions supported away from the spatial boundary. Direct conjugation gives

$$L = \theta P\theta^{-1} = i\partial_t - \Delta + 2\kappa\nabla\psi\cdot\nabla + \kappa\Delta\psi - \kappa^2|\nabla\psi|^2 - i\kappa'\psi.$$

Lemma 2.5. *Fix $T > 0$ and the phase (2.1). There exist $s_0 > 0$ and $C > 0$, depending only on T, U, and the fixed phase, such that, for every $s \geq s_0$,*

$$\int_0^T \int_U (\kappa|\nabla w|^2 + \kappa^3|w|^2)\, \mathrm{d}x\, \mathrm{d}t \leq C\int_0^T \int_U |Lw|^2\, \mathrm{d}x\, \mathrm{d}t \tag{2.11}$$

for every $w \in C_c^\infty((0,T) \times U; \mathbb{C})$. The same estimate holds for smooth functions with a fixed compact spatial support in U that are flat at both temporal endpoints. Here flatness means smooth extension to $[0,T] \times U$ with every temporal derivative equal to zero at $t = 0, T$.

Proof. Put $g = \nabla\psi$ and decompose $L = A + B$, where

$$A = i\partial_t - \Delta - \kappa^2|g|^2, \qquad B = 2\kappa g\cdot\nabla + \kappa\Delta\psi - i\kappa'\psi.$$

On the stated test class, integration by parts gives $A^* = A$ and $B^* = -B$. Thus

$$\|Lw\|_{L^2}^2 = \|Aw\|_{L^2}^2 + \|Bw\|_{L^2}^2 + 2\,\mathrm{Re}\int_{(0,T)\times U} Aw\,\overline{Bw}\, \mathrm{d}x\, \mathrm{d}t.$$

The L^2 norms in this proof are taken over $(0,T) \times U$. A direct calculation of the commutator gives

$$[A, B] = -8\kappa\Delta + 8\kappa^3|g|^2 + \kappa''\psi + 4i\kappa' g\cdot\nabla + 2i\kappa'\Delta\psi.$$

The spatial terms follow from

$$[-\Delta,\, 2\kappa g\cdot\nabla + \kappa\Delta\psi] = -8\kappa\Delta, \qquad [-\kappa^2|g|^2,\, 2\kappa g\cdot\nabla] = 8\kappa^3|g|^2,$$

because $D^2\psi = 2I$ and $\Delta\psi = 2n$. The remaining terms follow from

$$[i\partial_t,\, 2\kappa g\cdot\nabla + \kappa\Delta\psi] = 2i\kappa' g\cdot\nabla + i\kappa'\Delta\psi,$$

$$[-\Delta,\, -i\kappa'\psi] = 2i\kappa' g\cdot\nabla + i\kappa'\Delta\psi, \qquad [i\partial_t,\, -i\kappa'\psi] = \kappa''\psi.$$

Taking the real quadratic form of the commutator gives

$$\begin{aligned} 2\,\mathrm{Re}\int_{(0,T)\times U} Aw\,\overline{Bw}\,\mathrm{d}x\,\mathrm{d}t = 8\int_{(0,T)\times U}\kappa|\nabla w|^2\,\mathrm{d}x\,\mathrm{d}t \\ + 8\int_{(0,T)\times U}\kappa^3|g|^2|w|^2\,\mathrm{d}x\,\mathrm{d}t + \int_{(0,T)\times U}\kappa''\psi|w|^2\,\mathrm{d}x\,\mathrm{d}t \\ - 4\int_{(0,T)\times U}\kappa'\,\mathrm{Im}(g\cdot\nabla w\,\overline{w})\,\mathrm{d}x\,\mathrm{d}t. \end{aligned} \tag{2.12}$$

The support conditions eliminate the boundary terms, and $2i\kappa'\Delta\psi$ contributes nothing to the real quadratic form.

Let

$$d = \inf_U |g| > 0, \qquad G = \sup_U |g|, \qquad H = \sup_U |\psi|.$$

Young's inequality gives

$$4|\kappa'|\,G|\nabla w||w| \le 2\kappa|\nabla w|^2 + 2G^2\frac{|\kappa'|^2}{\kappa}|w|^2.$$

Lemma 2.1 implies

$$\frac{|\kappa'|^2}{\kappa} \le \frac{T^2}{s^2}\kappa^3, \qquad |\kappa''| \le \frac{2T^2}{s^2}\kappa^3.$$

Combining these bounds, we find that the right-hand side of (2.12) is bounded below by

$$6\int_{(0,T)\times U}\kappa|\nabla w|^2\,\mathrm{d}x\,\mathrm{d}t + \left(8d^2 - \frac{2T^2(G^2+H)}{s^2}\right)\int_{(0,T)\times U}\kappa^3|w|^2\,\mathrm{d}x\,\mathrm{d}t.$$

We choose s so that the coefficient in parentheses is at least $4d^2$. The decomposition of $\|Lw\|_{L^2}^2$ then gives (2.11).

To extend the estimate to flat functions, we multiply w by temporal cutoffs equal to one on $[2\varepsilon, T-2\varepsilon]$ and zero near the endpoints, with first derivatives of order $O(\varepsilon^{-1})$. Flatness makes the commutator with $i\partial_t$ tend to zero in L^2 and controls all polynomial factors in κ, κ', and κ'' above. The estimate therefore passes to the limit. □

To obtain a zeroth-order observation term, we regularize the test function before applying the strong estimate. The resulting source norm controls the spatial cutoff errors without differentiating the observed state. For this purpose, we use the spatial operator S_κ from Lemma 2.2.

Lemma 2.6. *Fix $T > 0$, a compact set $K \subset U$, and $\chi_2 \in C_c^\infty(U;[0,1])$ equal to one on a neighborhood of K. The spatial coefficients of L admit fixed smooth bounded extensions to $\mathbb{R}^n$, independent of s and t. Let $\widetilde{L}_s$ denote the resulting differential operator on $(0,T)\times\mathbb{R}^n$, chosen to agree with L near $(0,T)\times\operatorname{supp}\chi_2$. There exist $s_0 > 0$ and $C > 0$, depending only on the fixed spatial data and T, such that, for every $s \ge s_0$, the following estimate holds, with the subscript s on the operators suppressed:*

$$\int_0^T \kappa^3\|v(t)\|_{L^2(\mathbb{R}^n)}^2\,\mathrm{d}t \le C\int_0^T \|S_\kappa\widetilde{L}v(t)\|_{L^2(\mathbb{R}^n)}^2\,\mathrm{d}t \tag{2.13}$$

whenever $v \in C_c^\infty((0,T)\times\mathbb{R}^n;\mathbb{C})$ has spatial support in K. The estimate also holds when v is smooth with that spatial support and flat at $t = 0, T$.

Proof. We choose fixed functions $a_1, \dots, a_n, b, c, d \in C_b^\infty(\mathbb{R}^n)$ which agree, respectively, with

$$\partial_1\psi, \dots, \partial_n\psi, \quad \Delta\psi, \quad |\nabla\psi|^2, \quad \psi$$

on a neighborhood of $\operatorname{supp}\chi_2$. These extensions are independent of s and t. Set

$$\widetilde{L} = i\partial_t - \Delta + 2\kappa a\cdot\nabla + \kappa b - \kappa^2 c - i\kappa' d.$$

All spatial norms of the fixed coefficients are finite. We may increase the lower bound on s so that $\kappa(t) \geq 1$ throughout $(0,T)$.

For each fixed t, set $z = S_\kappa v$ and $w = \chi_2 z$. We use the energy

$$E_\kappa(z) = \kappa\|\nabla z\|^2_{L^2(\mathbb{R}^n)} + \kappa^3\|z\|^2_{L^2(\mathbb{R}^n)}.$$

Lemma 2.2 gives the exact identity

$$E_\kappa(S_\kappa v) = \kappa^3\|v\|^2_{L^2(\mathbb{R}^n)}. \tag{2.14}$$

Let $r = (1-\chi_2)z$ and $E = \operatorname{supp}\nabla\chi_2$. The distance from K to both $\operatorname{supp}(1-\chi_2)$ and E is positive. By Lemma 2.3, for every integer $N \geq 1$,

$$\begin{aligned}\|r\|_{L^2} + \|\nabla r\|_{L^2} + \|1_E z\|_{L^2} + \|1_E\nabla z\|_{L^2} \\ \leq C_N\kappa^{-N}\|v\|_{L^2}.\end{aligned} \tag{2.15}$$

The fixed separation makes these constants independent of t and s. The product rule also controls the terms in which a derivative falls on $1-\chi_2$. Consequently,

$$E_\kappa(r) \leq C_N\kappa^{3-2N}\|v\|^2_{L^2}.$$

Since $z = w + r$, (2.14) implies

$$\kappa^3\|v\|^2_{L^2} \leq 2E_\kappa(w) + C_N\kappa^{3-2N}\|v\|^2_{L^2}.$$

With $N = 2$ fixed, increasing $\inf_t \kappa(t)$ absorbs the last term. In particular,

$$\frac{1}{4}\kappa^3\|v\|^2_{L^2} \leq E_\kappa(w). \tag{2.16}$$

We next estimate the conjugated equation. Since $\widetilde{L} = L$ near $\operatorname{supp}\chi_2$, we have

$$Lw = \chi_2 S_\kappa \widetilde{L} v + \chi_2[\widetilde{L}, S_\kappa]v + [\widetilde{L}, \chi_2]S_\kappa v. \tag{2.17}$$

The first commutator is

$$\begin{aligned}[\widetilde{L}, S_\kappa]v = i(\partial_t S_\kappa)v + 2\kappa\sum_{j=1}^n [a_j, S_\kappa]\partial_j v + \kappa[b, S_\kappa]v \\ - \kappa^2[c, S_\kappa]v - i\kappa'[d, S_\kappa]v.\end{aligned}$$

Applying Lemma 2.2 to these terms gives

$$\|[\widetilde{L}, S_\kappa]v\|_{L^2} \leq C\left(\kappa + 1 + \frac{|\kappa'|}{\kappa}\right)\|v\|_{L^2}.$$

Here the time derivative contributes $\|\partial_t S_\kappa\|_{L^2\to L^2} \le C|\kappa'|/\kappa$. By Lemma 2.1, $|\kappa'|/\kappa \le (T/s)\kappa$. Hence, once $s \ge T$,

$$\|[\widetilde{L}, S_\kappa]v\|_{L^2} \le C\kappa\|v\|_{L^2}. \tag{2.18}$$

Thus the term arising from the singular time coefficient has the same order as the spatial commutator terms.

The second commutator has the explicit form

$$[\widetilde{L}, \chi_2]z = -2\nabla\chi_2 \cdot \nabla z - (\Delta\chi_2)z + 2\kappa(a \cdot \nabla\chi_2)z.$$

Its coefficients are supported away from K. The same off-diagonal estimate as in (2.15) (with $\operatorname{supp}\Delta\chi_2$ included, if necessary) gives

$$\|[\widetilde{L}, \chi_2]S_\kappa v\|_{L^2} \le C_N\kappa^{1-N}\|v\|_{L^2}. \tag{2.19}$$

By applying Lemma 2.5 to w and combining the energy comparison (2.16) with (2.17)–(2.19), we obtain

$$\int_0^T \kappa^3\|v\|_{L^2}^2 \,\mathrm{d}t \le C_1\int_0^T \|S_\kappa\widetilde{L}v\|_{L^2}^2 \,\mathrm{d}t + C_2\int_0^T \kappa^2\|v\|_{L^2}^2 \,\mathrm{d}t.$$

The constants C_1, C_2 are independent of s. Since $\inf_t \kappa(t) = 4s/T^2$, we can absorb the last term by choosing s so that $\inf_t \kappa(t) \ge 2C_2$. We obtain (2.13).

Finally, let v be flat at both endpoints. We apply the result to $v_\varepsilon = \rho_\varepsilon v$, with the temporal cutoffs used in Lemma 2.5. Since S_κ acts only in space,

$$S_\kappa\widetilde{L}v_\varepsilon = \rho_\varepsilon S_\kappa\widetilde{L}v + i\rho_\varepsilon' S_\kappa v.$$

The last term tends to zero in L^2 by $\|S_\kappa\|_{L^2\to L^2} \le 1$ and flatness of v. All remaining terms converge in the displayed norms, which proves the extension to flat functions. No bound on $\sup_t \kappa$ is needed. □

Proof of Proposition 2.4. The collar assumption allows us to choose a cutoff $\chi \in C_c^\infty(\Omega)$ such that

$$0 \le \chi \le 1, \qquad \chi = 1 \text{ on } \Omega \setminus \omega, \qquad \operatorname{supp}\nabla\chi \cup \operatorname{supp}\Delta\chi \subset \omega.$$

Choose χ_2 as in Lemma 2.6, with $K = \operatorname{supp}\chi$. Extend

$$v = \theta\chi p$$

by zero from Ω to $\mathbb{R}^n$. The compact support of χ makes this extension smooth in space. Since $\psi < -1$ and p is smooth up to $t = 0, T$, the extension is also flat at both temporal endpoints: every derivative is bounded by a polynomial in κ times a decaying exponential. Thus Lemma 2.6 applies.

On the support of v, the extended operator agrees with the conjugated operator. Hence

$$\widetilde{L}v = \theta P(\chi p) = \theta\chi Pp + \theta[P, \chi]p.$$

The first term satisfies, at every t,

$$\|S_\kappa(\theta\chi Pp)\|_{L^2(\mathbb{R}^n)} \le \|\theta Pp\|_{L^2(\Omega)}.$$

To estimate the cutoff term, we write its derivative in divergence form and apply the regularizer. With $q = \theta p$ in Ω,

$$\theta[P, \chi]p = -2\operatorname{div}(q\nabla\chi) + (\Delta\chi + 2\kappa\nabla\psi \cdot \nabla\chi)q. \tag{2.20}$$

The products on the right have compact support in ω and are extended by zero to $\mathbb{R}^n$. By Lemma 2.2,

$$\|S_\kappa \operatorname{div}(q\nabla\chi)\|_{L^2(\mathbb{R}^n)} \le C\kappa\|q\|_{L^2(\omega)}.$$

The multiplication term in (2.20) satisfies the same bound, since $\|S_\kappa\| \le 1$ and $\kappa \ge 1$. Thus

$$\|S_\kappa \widetilde{L} v\|^2_{L^2(\mathbb{R}^n)} \le C\left(\|\theta P p\|^2_{L^2(\Omega)} + \kappa^2\|\theta p\|^2_{L^2(\omega)}\right).$$

Inserting this bound into Lemma 2.6, we obtain

$$\int_Q \kappa^3\theta^2\chi^2|p|^2 \le C\left(\int_Q \theta^2|Pp|^2 + \int_{Q_\omega} \kappa^2\theta^2|p|^2\right).$$

Finally, $\chi = 1$ outside ω implies pointwise that $|p|^2 \le \chi^2|p|^2 + 1_\omega|p|^2$. Adding the resulting observation term and using $\kappa^2 \le \kappa^3$, we obtain (2.10). The argument requires no integration by parts at $\partial\Omega$. □

Remark 2.7. The zeroth-order left-hand side in (2.10) cannot in general be replaced by a left-hand side that also controls $\int_Q \kappa\theta^2|\nabla p|^2$, while keeping the same zeroth-order source and observation terms. To see this, fix s and let e_j be normalized Dirichlet eigenfunctions, $-\Delta e_j = \lambda_j e_j$, with $\lambda_j \to \infty$. Then $p_j(t,x) = e^{i\lambda_j t}e_j(x)$ satisfies $Pp_j = 0$. On $J = [T/4, 3T/4]$ the minimum of $\kappa\theta^2$ over $J \times \overline{\Omega}$ is a positive number a_s. Therefore

$$\int_Q \kappa\theta^2|\nabla p_j|^2 \ge a_s|J|\,\lambda_j.$$

In contrast, if $b_s = \sup_Q \kappa^3\theta^2 < \infty$, then

$$\int_{Q_\omega} \kappa^3\theta^2|p_j|^2 \le b_s T.$$

This contradicts such a strengthened estimate as $j \to \infty$. Lemma 2.5 remains valid, since its test functions have compact spatial support. The order reduction is therefore essential to obtaining an observation term without spatial derivatives.

2.3 A local boundary phase. We next establish the estimate under the Euclidean local phase condition of Assumption 1.2. We use a Dirichlet regularizer so that the spatial localization can reach the unobserved boundary. The interval case of Theorem 1.4 is treated separately in Section 2.4.

Assume (1.3), and retain the notation K_ω and $\mathcal{O}$ from the introduction. We subtract a constant from ψ so that $\psi < -1$ on $\overline{\Omega}$. All supports and neighborhoods in this subsection are relative to $\overline{\Omega}$.

Proposition 2.8. *Assume (1.3), and fix a constant shift of ψ such that $\psi < -1$ on $\overline{\Omega}$. For every $T > 0$, set $\kappa = s/[t(T-t)]$ and $\theta = e^{\kappa\psi}$. There exist $s_0 > 0$ and $C > 0$, depending only on T and the fixed spatial data, such that*

$$\int_Q \kappa^3\theta^2|p|^2 \le C\left(\int_Q \theta^2|Pp|^2 + \int_{Q_\omega} \kappa^3\theta^2|p|^2\right)$$

for every $s \ge s_0$ and every $p \in C^\infty([0,T] \times \overline{\Omega}; \mathbb{C})$ satisfying $p = 0$ on Σ. No condition is imposed on $p(0)$ or $p(T)$. Here the fixed spatial data comprise the domain, control region, phase, phase neighborhood, and the constants in (1.3). In particular, s_0 and C are independent of s and p.

Tangency enters the proof in two places. It cancels the boundary term in the strong estimate and ensures that the transport field preserves the Dirichlet boundary condition in the resolvent commutator below. A favorable boundary sign alone does not ensure the latter property. Thus the normal condition in Assumption 1.2 is a restriction of this order-reduction argument, not a necessary condition for exact controllability.

Put $A = -\Delta_D$ on $H = L^2(\Omega)$ and

$$R_\mu = (\mu + A)^{-1}, \qquad S_\kappa = \kappa(\kappa^2 + A)^{-1/2}.$$

In this subsection, S_κ denotes this Dirichlet operator. Its functional-calculus representation is

$$S_\kappa = \frac{\kappa}{\pi} \int_0^\infty r^{-1/2} R_{\kappa^2 + r} \, \mathrm{d}r. \tag{2.21}$$

The integral converges in operator norm on H.

Lemma 2.9. *Let $b \in C^\infty(\overline{\Omega})$, and let $X = a \cdot \nabla$, where $a \in C^\infty(\overline{\Omega}; \mathbb{R}^n)$ satisfies $a \cdot \nu = 0$ on $\partial\Omega$. For every $\kappa \geq 1$ and $F \in L^2(\Omega; \mathbb{C}^n)$, with* div F *defined in the conjugate-linear dual of $H_0^1(\Omega; \mathbb{C})$, one has*

$$\begin{aligned} &\|S_\kappa\|_{H \to H} \leq 1, \qquad \|S_\kappa \operatorname{div} F\|_H \leq \kappa \|F\|_{L^2(\Omega)^n}, \\ &\|[b, S_\kappa]\|_{H \to H} \leq C\kappa^{-1}, \qquad \|[X, S_\kappa]\|_{H \to H} \leq C, \\ &\|\partial_t S_{\kappa(t)}\|_{H \to H} \leq C|\kappa'|/\kappa. \end{aligned}$$

The commutator with X, initially defined on smooth Dirichlet functions, extends to the displayed bounded operator. For every $v \in H$, one also has

$$\kappa \|\nabla S_\kappa v\|_2^2 + \kappa^3 \|S_\kappa v\|_H^2 = \kappa^3 \|v\|_H^2. \tag{2.22}$$

The constants depend only on the fixed domain and coefficients b, a, and are independent of κ, v, and F.

Proof. The contraction and time derivative estimates, together with (2.22), follow from the spectral theorem. The divergence estimate follows by duality from

$$\|\nabla S_\kappa u\|_2^2 = \sum_j \frac{\kappa^2 \lambda_j}{\kappa^2 + \lambda_j} |(u, e_j)_H|^2 \leq \kappa^2 \|u\|_H^2.$$

Here the divergence is interpreted in $H^{-1}(\Omega)$, the dual of $H_0^1(\Omega)$. Thus no boundary flux of F is introduced.

For $\mu \geq 1$ the spectral theorem and duality give

$$\|R_\mu\|_{H \to H} \leq \mu^{-1}, \qquad \|R_\mu\|_{H \to H_0^1} + \|R_\mu\|_{H^{-1} \to H} \leq C\mu^{-1/2}.$$

Tangency ensures that X preserves the zero boundary trace. In particular, for $v \in H$, we have $R_\mu v \in D(A)$ and $X R_\mu v \in H_0^1$, because tangential derivatives of a Dirichlet function have zero trace. Interpreting $X : H \to H^{-1}$ distributionally, we obtain the weak resolvent identity

$$[X, R_\mu] = R_\mu [A, X] R_\mu,$$

where

$$[A, X]u = -2 \sum_{i,j} \partial_i ((\partial_i a_j) \partial_j u) + \sum_j (\Delta a_j) \partial_j u.$$

In this divergence form, the commutator defines a bounded map $[A, X] : H_0^1 \to H^{-1}$, whose norm depends only on fixed derivatives of a. To verify the resolvent identity, note that $(\mu + A)XR_\mu v = Xv + [A, X]R_\mu v$ in H^{-1}; applying its Dirichlet resolvent yields the identity. This argument does not require X to preserve $D(A)$. Consequently,

$$\|[X, R_\mu]\|_{H\to H} \le C\mu^{-1}.$$

Similarly,

$$[b, R_\mu] = R_\mu[A, b]R_\mu, \qquad [A, b] = -2\nabla b \cdot \nabla - \Delta b,$$

so that

$$\|[b, R_\mu]\|_{H\to H} \le C\mu^{-3/2}.$$

Integration in (2.21), using

$$\int_0^\infty \frac{r^{-1/2}}{\kappa^2 + r}\,\mathrm{d}r = \frac{\pi}{\kappa}, \qquad \int_0^\infty \frac{r^{-1/2}}{(\kappa^2 + r)^{3/2}}\,\mathrm{d}r = \frac{2}{\kappa^2},$$

gives the two commutator bounds. We first justify interchanging the integral and the unbounded operator X for $v \in D(A)$. The truncated resolvent integrals converge in the graph norm of $D(A)$, because A commutes with R_μ. Thus X can be applied before passing to the limit. The integrals of the commutators converge in operator norm on H, and the resulting commutator extends to H by density. □

Lemma 2.10. *Let $K, E \subset \overline{\Omega}$ be closed sets with* $\operatorname{dist}(K, E) > 0$. *For every integer $N \ge 1$, there exists $C_N > 0$ such that, for all $v \in H$ supported in K and all $\kappa \ge 1$,*

$$\|1_E S_\kappa v\|_H + \|1_E \nabla S_\kappa v\|_2 \le C_N \kappa^{-N} \|v\|_H.$$

The same estimate holds for $\|\eta S_\kappa v\|_{H^1}$ when η is a fixed smooth cutoff supported away from K. The constants depend only on the domain, the fixed sets, N, and, in the cutoff version, η.

Proof. Let $u = R_\mu v$, with $\mu \ge 1$. A real cutoff $\eta \in C^\infty(\overline{\Omega})$ that vanishes near K is admissible in the Dirichlet energy identity, even when its support meets the boundary. Testing $(\mu - \Delta)u = v$ with $\eta^2 u$ and taking the real part gives

$$\mu\|\eta u\|_H^2 + \|\nabla(\eta u)\|_2^2 = \||\nabla\eta| u\|_H^2. \tag{2.23}$$

We iterate (2.23) using finitely many nested cutoff neighborhoods that separate E from K. Each step gains a factor $\mu^{-1/2}$ in the localized L^2 norm. Starting from $\|u\|_H \le \mu^{-1}\|v\|_H$, iteration gives, for any integer $m \ge 1$,

$$\|1_E R_\mu v\|_H \le C_m \mu^{-1-m/2}\|v\|_H, \qquad \|1_E \nabla R_\mu v\|_2 \le C_m \mu^{-1/2-m/2}\|v\|_H.$$

To obtain the gradient estimate, we use the gradient term in (2.23) at the last step; the preceding $m - 1$ steps use the L^2 term. The constants can depend on m and the separation, which are fixed independently of μ.

We insert these estimates into (2.21). The substitution $r = \kappa^2 z$ gives

$$\|1_E S_\kappa v\|_H \le C_m \kappa^{-m}\|v\|_H, \qquad \|1_E \nabla S_\kappa v\|_2 \le C_m \kappa^{1-m}\|v\|_H.$$

The integrals converge for $m \ge 1$. Taking $m = N + 1$ proves the claim. The product rule proves the cutoff version. □

Proof of Proposition 2.8. If $K_\omega = \varnothing$, then $\omega = \Omega$ and the estimate follows from its observation term. We henceforth assume $K_\omega \neq \varnothing$. Write $g = \nabla\psi$ and

$$L = \theta P \theta^{-1} = i\partial_t - \Delta + 2\kappa g \cdot \nabla + \kappa\Delta\psi - \kappa^2|g|^2 - i\kappa'\psi.$$

We first consider a smooth Dirichlet function w whose spatial support is a compact subset of $\mathcal{O}$ and which is flat at both temporal endpoints. Decompose $L = \mathcal{A} + \mathcal{B}$, where

$$\mathcal{A} = i\partial_t - \Delta - \kappa^2|g|^2, \qquad \mathcal{B} = 2\kappa g \cdot \nabla + \kappa\Delta\psi - i\kappa'\psi.$$

Integration by parts gives

$$\begin{aligned} 2\,\mathrm{Re}\int_Q \mathcal{A}w\,\overline{\mathcal{B}w} = {} & 4\int_Q \kappa\, D^2\psi(\nabla w, \nabla\overline{w}) + 4\int_Q \kappa^3 D^2\psi(g,g)|w|^2 \\ & - \int_Q \kappa\Delta^2\psi\,|w|^2 + \int_Q \kappa''\psi|w|^2 - 4\int_Q \kappa'\,\mathrm{Im}(g\cdot\nabla w\,\overline{w}) \\ & - 2\int_0^T\int_{\partial\Omega} \kappa\partial_\nu\psi\,|\partial_\nu w|^2\,\mathrm{d}\sigma\,\mathrm{d}t. \end{aligned}$$

The boundary integral vanishes because $\partial_\nu\psi = 0$ on $\mathcal{O}\cap\partial\Omega$ and w vanishes near the rest of the boundary. Strict convexity on $\mathcal{O}$ and the lower bound on $|g|$ make the first two volume terms coercive. More precisely, Young's inequality and Lemma 2.1 give constants $c, C > 0$, depending only on the spatial data, such that

$$2\,\mathrm{Re}(\mathcal{A}w, \mathcal{B}w)_{L^2(Q)} \geq c\int_Q \kappa|\nabla w|^2 + \int_Q \left(c - C\frac{T^2}{s^2} - C\kappa^{-2}\right)\kappa^3|w|^2.$$

Here the last term accounts for $-\kappa\Delta^2\psi$. We therefore take T/s sufficiently small and $\inf_t \kappa(t)$ sufficiently large to obtain

$$\int_0^T \left(\kappa\|\nabla w\|_2^2 + \kappa^3\|w\|_H^2\right)\,\mathrm{d}t \leq C\|Lw\|_{L^2(Q)}^2. \tag{2.24}$$

We also need the estimate for functions in $L^2(0,T;D(A))\cap H^1(0,T;H)$ with compact temporal support and fixed spatial support compactly contained in $\mathcal{O}$. We first apply Dirichlet spectral truncation and time smoothing. We then multiply the approximants by a fixed smooth cutoff supported in $\mathcal{O}$ and equal to one near the original support. The graph norm of $D(A)$ is equivalent to the H^2 norm, so this multiplication preserves both convergences. Since the coefficients of L are bounded on the temporal support, the estimate passes to the limit. The estimate also extends to functions in $C^\infty([0,T];D(A))$ that are flat at both endpoints in the graph norm. We insert temporal cutoffs and then remove them: flatness dominates every polynomial in the time weights, and the error $i\rho_\varepsilon' w$ tends to zero in $L^2(Q)$.

We choose $\chi, \chi_2, \rho \in C^\infty(\overline{\Omega})$, taking values in $[0,1]$, with successively nested supports in $\mathcal{O}$. We require that $\chi = 1$ on a neighborhood of K_ω, that $\chi_2 = 1$ on a neighborhood of $\mathrm{supp}\,\chi$, and that $\rho = 1$ on a neighborhood of $\mathrm{supp}\,\chi_2$. In particular,

$$\chi = 1 \text{ on } \Omega\setminus\omega, \qquad (\mathrm{supp}\,\nabla\chi \cup \mathrm{supp}\,\Delta\chi)\cap\Omega \subset \omega.$$

Such cutoffs exist because K_ω is compact and is contained in the relatively open set $\mathcal{O}$. Put

$$a = \rho\nabla\psi, \qquad \widetilde{L} = i\partial_t - \Delta + 2\kappa a\cdot\nabla + \kappa\Delta\psi - \kappa^2|\nabla\psi|^2 - i\kappa'\psi.$$

Then $a \cdot \nu = 0$ on the entire boundary, since the field is tangent on $\mathcal{O} \cap \partial\Omega$ and vanishes on the rest. Moreover, $\widetilde{L} = L$ near $\operatorname{supp} \chi_2$.

Let $v \in C^\infty([0,T]; D(A))$ be supported spatially in $K = \operatorname{supp} \chi$ and flat at both temporal endpoints in the graph norm of $D(A)$. Set $z = S_\kappa v$ and $w = \chi_2 z$. Both functions are smooth and flat at the temporal endpoints in the graph norm. Indeed, spectral calculus gives

$$\|\partial_\kappa^r S_\kappa\|_{D(A)\to D(A)} \le C_r \kappa^{-r} \qquad (r = 0, 1, 2, \ldots).$$

Time differentiation introduces only polynomial factors in κ and its derivatives, which are dominated by the flatness of v. Multiplication by χ_2 is bounded on $D(A)$. Lemma 2.10 and (2.22) imply

$$\tfrac{1}{4}\kappa^3 \|v\|_H^2 \le \kappa \|\nabla w\|_2^2 + \kappa^3 \|w\|_H^2$$

for sufficiently large $\inf_t \kappa$. In fact the energy of $(1-\chi_2)z$ is at most $C_N \kappa^{3-2N} \|v\|_H^2$. We have the identity

$$Lw = \chi_2 S_\kappa \widetilde{L} v + \chi_2 [\widetilde{L}, S_\kappa] v + [\widetilde{L}, \chi_2] S_\kappa v.$$

Lemma 2.9 gives

$$\|[\widetilde{L}, S_\kappa] v\|_H \le C \left(\kappa + 1 + \frac{|\kappa'|}{\kappa} \right) \|v\|_H \le C\kappa \|v\|_H,$$

once $s \ge T$ and $\kappa \ge 1$. The tangent transport and the multiplier $-\kappa^2 |\nabla\psi|^2$ each contribute $\mathcal{O}(\kappa)$; the multiplier $-i\kappa'\psi$ contributes $\mathcal{O}(|\kappa'|/\kappa)$. The cutoff commutator is supported a positive distance from K, so Lemma 2.10 gives

$$\|[\widetilde{L}, \chi_2] S_\kappa v\|_H \le C_N \kappa^{1-N} \|v\|_H.$$

By applying (2.24) to w and absorbing the resulting $C \int_0^T \kappa^2 \|v\|_H^2 \, \mathrm{d}t$, we obtain

$$\int_0^T \kappa^3 \|v\|_H^2 \, \mathrm{d}t \le C \int_0^T \|S_\kappa \widetilde{L} v\|_H^2 \, \mathrm{d}t. \tag{2.25}$$

Take $v = \theta \chi p$. It has zero Dirichlet trace and is flat at both temporal endpoints. On its support,

$$\widetilde{L} v = \theta \chi P p + \theta [P, \chi] p.$$

With $q = \theta p$, the last term is

$$\theta [P, \chi] p = -2 \operatorname{div}(q \nabla \chi) + (\Delta \chi + 2\kappa \nabla\psi \cdot \nabla\chi) q.$$

The contraction and divergence bounds of Lemma 2.9 imply

$$\|S_\kappa \widetilde{L} v\|_H^2 \le C \left(\|\theta P p\|_H^2 + \kappa^2 \|\theta p\|_{L^2(\omega)}^2 \right).$$

Inserting this estimate into (2.25) and using $|p|^2 \le \chi^2 |p|^2 + 1_\omega |p|^2$, we obtain the Carleman inequality. Together with the strict negativity of ψ, this inequality supplies the hypotheses for the weighted control construction in Section 3.1 and the nonlinear argument in Section 3.2, with $\vartheta = \theta$. □

2.4 One dimension case. On an interval, reflection allows us to treat any nonempty open control set. We apply the full-space estimate to two reflected pieces and fold the resulting weights back onto the interval. We use separate state and observation weights to avoid comparing exponential weights across the partition of unity.

Proposition 2.11. *Let $\Omega = (0, \ell)$, where $\ell > 0$, and let $\omega \subset (0, \ell)$ be a nonempty open set. Fix $T > 0$ and set $\kappa = s/[t(T - t)]$. There are positive weights W_s, V_s on Q, constants $\alpha, \beta, C > 0$, and $s_0 > 0$ such that*

$$\frac{1}{4} e^{-2\beta\kappa} \leq W_s \leq V_s \leq e^{-2\alpha\kappa} \tag{2.26}$$

and

$$\int_Q \kappa^3 W_s |p|^2 \leq C \left(\int_Q W_s |Pp|^2 + \int_{Q_\omega} \kappa^3 V_s |p|^2 \right) \tag{2.27}$$

for every $s \geq s_0$ and every $p \in C^\infty([0, T] \times [0, \ell]; \mathbb{C})$ satisfying $p(t, 0) = p(t, \ell) = 0$. The constants depend only on T, the interval, and a fixed subinterval compactly contained in ω, together with the associated cutoffs and phase. They are independent of s and p. No temporal endpoint condition is imposed on p.

Proof. We choose $0 < a < b < \ell$ such that $[a, b] \subset \omega$. We take $\chi_- \in C^\infty([0, \ell]; [0, 1])$ with

$$\chi_- = 1 \quad \text{on } [0, a], \qquad \chi_- = 0 \quad \text{on } [b, \ell],$$

and set $\chi_+ = 1 - \chi_-$. The cutoffs localize near opposite endpoints, and their nonzero derivatives are supported in $[a, b] \subset \omega$. In particular,

$$\chi_-^2 + \chi_+^2 \geq \frac{1}{2}. \tag{2.28}$$

We reflect the two localized functions across the left and right endpoints, respectively. The reflection maps are $R_- x = -x$ and $R_+ x = 2\ell - x$. We define functions q_-, q_+ on the real line by

$$q_-(x) = \begin{cases} \chi_-(x) p(x), & 0 \leq x \leq \ell, \\ -\chi_-(-x) p(-x), & -\ell \leq x < 0, \\ 0, & \text{otherwise}, \end{cases}$$

and

$$q_+(x) = \begin{cases} \chi_+(x) p(x), & 0 \leq x \leq \ell, \\ -\chi_+(2\ell - x) p(2\ell - x), & \ell < x \leq 2\ell, \\ 0, & \text{otherwise}. \end{cases}$$

The time variable is suppressed in these formulas. The supports of these functions are contained in $[-b, b]$ and $[a, 2\ell - a]$, respectively. Both functions depend smoothly on time with values in $H^2(\mathbb{R})$. At the reflecting point, the Dirichlet condition makes the odd extension continuous, and odd reflection also matches its first derivative. The distributional second derivative therefore contains no interface measure. This H^2 regularity is sufficient; higher smoothness across the interface is unnecessary.

We choose a bounded interval U containing $[-\ell, 2\ell]$ in its interior. We then choose $x_0 \notin \overline{U}$ and M so large that

$$\psi(x) = |x - x_0|^2 - M < -1 \quad \text{on } \overline{U}.$$

Write $\vartheta_0(t,x) = e^{\kappa(t)\psi(x)}$ and

$$\alpha = \min_{\overline{U}}(-\psi) > 0, \qquad \beta = \max_{\overline{U}}(-\psi).$$

We choose a fixed compact set $K' \Subset U$ whose interior contains both reflected supports, and fix the cutoff and coefficient extensions in Lemma 2.6 for K'. Set $v_j = \vartheta_0 q_j$. After applying a smooth temporal cutoff supported in $(0,T)$, we mollify in space. The mollified functions converge in $H^1_t L^2_x \cap L^2_t H^2_x$. All sufficiently small mollifications have support in K', so the constant in the lowering estimate is unchanged. The coefficients of L are bounded on the temporal support; hence the images under L converge in L^2, as do their images under $S_\kappa L$. Thus the lowering estimate extends to the present H^2 functions. For fixed (T,s), v_j is flat at both temporal endpoints in $H^2(\mathbb{R})$. We can therefore remove the temporal cutoff using $\|S_\kappa\|_{2\to 2} \le 1$ and exponential flatness, with the same estimate constant. We obtain

$$\int_0^T \kappa^3 \|\vartheta_0 q_j\|^2_{L^2(\mathbb{R})}\, \mathrm{d}t \le C \int_0^T \|S_\kappa(\vartheta_0 P q_j)\|^2_{L^2(\mathbb{R})}\, \mathrm{d}t, \qquad j \in \{-,+\}. \tag{2.29}$$

We next estimate the cutoff commutators. Let $\mathcal{O}_j$ denote odd reflection of a function on $(0,\ell)$ across the corresponding endpoint, followed by zero extension. This notation is applied below only to functions vanishing near the opposite endpoint. Define F_j, G_j on $(0,\ell)$ by

$$F_j = \chi_j' p, \qquad G_j = \chi_j'' p.$$

Extend F_j evenly and G_j oddly across the reflecting endpoint, and extend them by zero elsewhere. These extensions are smooth because χ_j' and χ_j'' are supported away from both endpoints. Direct differentiation gives

$$P q_j = \mathcal{O}_j(\chi_j P p) - 2\partial_x F_j + G_j.$$

On the original interval, the last two terms equal $-2\chi_j' p_x - \chi_j'' p$. On the reflected interval, differentiating the even extension of F_j gives the sign required by odd reflection. Thus the identity holds in L^2 and in distributions. Although $\mathcal{O}_j(\chi_j P p)$ may jump at the reflecting point, continuity of q_j and its first derivative excludes an interface measure. No boundary condition on Pp is needed.

Multiplication by the exponential weight yields

$$\vartheta_0(-2\partial_x F_j + G_j) = -2\partial_x(\vartheta_0 F_j) + 2\kappa\psi'\vartheta_0 F_j + \vartheta_0 G_j.$$

Since $\|S_\kappa\| \le 1$ and $\|S_\kappa \partial_x\| \le \kappa$, we can estimate the commutator in terms of p, without any derivative of p. We now fold the full-space integrals back onto the interval. For this purpose, introduce the weights on $(0,\ell)$,

$$w_-(t,x) = \vartheta_0(t,x)^2 + \vartheta_0(t,-x)^2, \qquad w_+(t,x) = \vartheta_0(t,x)^2 + \vartheta_0(t,2\ell - x)^2.$$

All original and reflected arguments lie in U. Thus the bounds determined by α and β apply to every exponential factor. At each time,

$$\|\vartheta_0 q_j\|_2^2 = \int_0^\ell \chi_j^2 w_j |p|^2\, \mathrm{d}x, \qquad \|\vartheta_0 \mathcal{O}_j(\chi_j P p)\|_2^2 = \int_0^\ell \chi_j^2 w_j |Pp|^2\, \mathrm{d}x.$$

Moreover,

$$\|S_\kappa[\vartheta_0(-2\partial_x F_j + G_j)]\|_2^2 \le C\kappa^2 \int_\omega w_j |p|^2\, \mathrm{d}x$$

for $\kappa \geq 1$. Inserting these estimates in (2.29) gives

$$\int_Q \kappa^3 \chi_j^2 w_j |p|^2 \leq C \left(\int_Q \chi_j^2 w_j |Pp|^2 + \int_{Q_\omega} \kappa^2 w_j |p|^2 \right).$$

Finally, set

$$W_s = \frac{1}{4}(\chi_-^2 w_- + \chi_+^2 w_+), \qquad V_s = \frac{1}{4}(w_- + w_+).$$

Adding the two estimates, dividing by four, and using $\kappa^2 \leq \kappa^3$ gives (2.27). Each w_j lies between $2e^{-2\beta\kappa}$ and $2e^{-2\alpha\kappa}$. Together with (2.28), this proves (2.26). In particular, $\theta = W_s^{1/2}$ and $\vartheta = V_s^{1/2}$ satisfy (3.1) with $c_* = 1/2$. For fixed s, these weights and their inverses are bounded on compact subsets of Q. □

The reflection construction applies only to linear test functions and therefore imposes no parity assumption on the nonlinearity. The interval case of Theorem 1.4 follows by using $\theta = W_s^{1/2}$ and $\vartheta = V_s^{1/2}$ in the control construction of Section 3.

Proposition 2.12. *Fix either the Euclidean local-phase geometry of Assumption 1.2 or an interval with any nonempty open control set, together with the spatial phase and cutoffs. There are constants $a, b, C_{\mathrm{Car}}, c_*, \alpha, \beta > 0$, depending only on these spatial data, with the following property. For every $T > 0$, set $\kappa(t) = s/[t(T-t)]$. If*

$$s \geq s_0(T) := aT + bT^2, \tag{2.30}$$

there are positive weights θ, ϑ satisfying the envelopes (3.1) and the Carleman estimate (3.2). The constants are independent of T and s. In particular, the threshold has order T as $T \downarrow 0$; no additional condition $s \geq 1$ is imposed. On an interval use $\theta = W_s^{1/2}$ and $\vartheta = V_s^{1/2}$.

Proof. We keep all spatial data fixed and track the two dimensionless quantities that enter the preceding proofs:

$$\delta = \frac{T}{s}, \qquad m = \inf_t \kappa(t) = \frac{4s}{T^2}.$$

We show that all preceding estimates hold once $\delta \leq \delta_0$ and $m \geq M_0$, for suitably chosen constants $\delta_0 > 0$ and $M_0 \geq 1$ depending only on the spatial data. By Lemma 2.1,

$$|\kappa'| \leq \delta \kappa^2, \qquad |\kappa''| \leq 2\delta^2 \kappa^3, \qquad |\kappa'|^2/\kappa \leq \delta^2 \kappa^3.$$

In the strong estimate, the mixed time term is at most $\varepsilon\kappa|\nabla w|^2 + C_\varepsilon \delta^2 \kappa^3 |w|^2$. The κ'' term has the same order. We choose $\delta_0 \leq 1$ sufficiently small to absorb these remainders, with a coercivity constant depending only on the phase. For a nonquadratic phase, the additional $\kappa\Delta^2\psi$ term is absorbed by a fixed lower bound on m. The boundary flux vanishes by tangency.

The spatial regularizer and off-diagonal estimates are uniform for $\kappa \geq 1$. Their time-dependent commutators satisfy

$$\|\partial_t S_{\kappa(t)}\|_{2\to 2} + \|\kappa'[\psi, S_\kappa]\|_{2\to 2} \leq C\frac{|\kappa'|}{\kappa} \leq C\delta_0 \kappa.$$

Every other regularizer commutator has norm at most $C\kappa$. Thus, with constants now independent of T, the lowering argument gives the remainder $C_2 \int_0^T \int_D \kappa^2 |v|^2 \, \mathrm{d}x \, \mathrm{d}t$, where $D = \mathbb{R}^n$ in the full-space argument and $D = \Omega$ in the Dirichlet argument (with $D = \mathbb{R}$ for the reflected interval). This term is absorbed when $m \geq 2C_2$. The localized energy comparison and the cutoff source

estimates require only further fixed lower bounds on m. We include all these bounds in M_0. Reflection and folding on an interval involve only fixed spatial maps and introduce no additional time dependence.

The temporal cutoff and density arguments do not introduce further thresholds. For each fixed positive pair (T, s), flatness ensures that the approximation errors tend to zero with the same estimate constants. Hence the estimates remain valid even when $s < 1$, provided $T/s \le \delta_0$ and $4s/T^2 \ge M_0$. By taking $a \ge \delta_0^{-1}$ and $b \ge M_0/4$, we obtain (2.30). The weight envelopes depend only on the extrema of the spatial phases. □

2.5 The Laplace–Beltrami operator.

Proof of Proposition 1.11. We extend Proposition 2.8 to the Riemannian setting, with constants independent of the time horizon. All derivatives and inner products below are taken with respect to g. Put $P_g = i\partial_t - \Delta_g$ and let $A = -\Delta_g$ have its Dirichlet realization when $\partial M \neq \varnothing$. Otherwise A is the self-adjoint Laplacian on M. Its group e^{itA} is unitary on $L^2(M)$ in both cases. If $K_\omega = \varnothing$, the Carleman estimate below follows from its observation term. We may therefore assume $K_\omega \neq \varnothing$. Shift ψ so that $\psi < -1$, and set $\kappa = s/[t(T-t)]$, $\theta = e^{\kappa\psi}$ and $V = \nabla_g \psi$. Then

$$\theta P_g \theta^{-1} = \mathcal{A} + \mathcal{B}, \qquad \mathcal{A} = i\partial_t - \Delta_g - \kappa^2 |V|_g^2, \quad \mathcal{B} = 2\kappa V \cdot \nabla_g + \kappa \Delta_g \psi - i\kappa' \psi.$$

For a smooth function w with spatial support compactly contained in $\mathcal{O}$, flat at the temporal endpoints and satisfying the Dirichlet condition where applicable, integration by parts gives

$$\begin{aligned} 2\,\mathrm{Re}(\mathcal{A}w, \mathcal{B}w)_{L^2} = {} & 4\int_0^T\!\!\int_M \kappa\, \mathrm{Hess}_g\, \psi(\nabla_g w, \nabla_g \bar{w}) + 4\int_0^T\!\!\int_M \kappa^3\, \mathrm{Hess}_g\, \psi(V,V)|w|^2 \\ & - \int_0^T\!\!\int_M \kappa \Delta_g^2 \psi |w|^2 + \int_0^T\!\!\int_M \kappa'' \psi |w|^2 - 4\int_0^T\!\!\int_M \kappa'\, \mathrm{Im}(V \cdot \nabla_g w\, \bar{w}) \\ & - 2\int_0^T\!\!\int_{\partial M} \kappa \partial_{\nu_g} \psi\, |\partial_{\nu_g} w|^2. \end{aligned}$$

Here and below, volume and boundary integrals are taken with respect to $\mathrm{d}V_g\, \mathrm{d}t$ and $\mathrm{d}S_g\, \mathrm{d}t$, respectively. The boundary term is absent on a closed manifold. No separate curvature term occurs in this scalar identity. Indeed,

$$[-\Delta_g, 2V \cdot \nabla_g + \Delta_g \psi] = -4\, \mathrm{div}_g(\mathrm{Hess}_g\, \psi\, \nabla_g \cdot) - \Delta_g^2 \psi,$$

where the Hessian is viewed as a symmetric endomorphism. This divergence form incorporates the curvature terms produced by commuting covariant derivatives.

The boundary term vanishes by tangency. Using $|\kappa'| \le (T/s)\kappa^2$ and $|\kappa''| \le 2(T/s)^2\kappa^3$, we obtain

$$\int_0^T (\kappa \|\nabla_g w\|_2^2 + \kappa^3 \|w\|_2^2)\, \mathrm{d}t \le C \|\theta P_g \theta^{-1} w\|_{L^2}^2 \tag{2.31}$$

once $T/s \le \delta_0$ and $\inf_t \kappa \ge M_0$, with constants depending only on the fixed geometric data.

The order reduction also has an intrinsic form. For $\mu \ge 1$ and $\kappa \ge 1$, set $R_\mu = (\mu + A)^{-1}$ and $S_\kappa = \kappa(\kappa^2 + A)^{-1/2}$. For a smooth vector field X tangent to ∂M, let E_X denote the symmetric part of its covariant derivative. The identity

$$[A, X]u = -2\, \mathrm{div}_g(E_X \nabla_g u) + \langle \nabla_g\, \mathrm{div}_g\, X, \nabla_g u \rangle_g$$

defines a bounded map $H_0^1(M) \to H^{-1}(M)$. On a closed manifold these spaces are $H^1(M)$, with its full norm, and its dual. Tangency ensures that $XR_\mu u$ has zero boundary trace. Consequently, the weak resolvent identity $[X, R_\mu] = R_\mu[A, X]R_\mu$ gives $\|[X, R_\mu]\|_{2\to2} \le C\mu^{-1}$. For a smooth multiplier b, $[A, b] = -2\nabla_g b \cdot \nabla_g - \Delta_g b$ similarly gives $\|[b, R_\mu]\|_{2\to2} \le C\mu^{-3/2}$. Integrating the resolvents as in (2.21) yields

$$\|[X, S_\kappa]\|_{2\to2} \le C, \qquad \|[b, S_\kappa]\|_{2\to2} \le C\kappa^{-1}, \qquad \|S_\kappa \operatorname{div}_g F\|_2 \le \kappa\|F\|_2.$$

The last bound follows from spectral calculus and duality, which also give

$$\kappa\|\nabla_g S_\kappa v\|_2^2 + \kappa^3\|S_\kappa v\|_2^2 = \kappa^3\|v\|_2^2, \qquad \|\partial_t S_\kappa\|_{2\to2} \le C|\kappa'|/\kappa.$$

These spectral estimates use only $A \ge 0$ and also hold on $\ker A$ in the closed case. The same cutoff argument gives the off-diagonal estimates of Lemma 2.10. Indeed, if $u = R_\mu v$ and η vanishes near $\operatorname{supp} v$,

$$\mu\|\eta u\|_2^2 + \|\nabla_g(\eta u)\|_2^2 = \||\nabla_g \eta|_g u\|_2^2.$$

We choose the nested cutoffs χ, χ_2, ρ as in the proof of Proposition 2.8, and replace V in the transport term by the globally tangent field ρV. We apply (2.31) to $\chi_2 S_\kappa v$, where $v = \theta\chi p$. The preceding commutator bounds give a remainder $C\int_0^T \kappa^2\|v\|_2^2 \,\mathrm{d}t$, which we absorb by increasing M_0. For the localized source, we use

$$\theta[P_g, \chi]p = -2\operatorname{div}_g(\theta p\nabla_g\chi) + (\Delta_g\chi + 2\kappa\langle V, \nabla_g\chi\rangle_g)\theta p.$$

The cutoff derivatives in this identity are supported in ω. The density and temporal cutoff arguments of the Euclidean proof apply through the spectral resolution of A and elliptic regularity. Thus, for every smooth test function p with unrestricted temporal endpoint values and homogeneous Dirichlet trace when $\partial M \ne \varnothing$,

$$\int_0^T\int_M \kappa^3\theta^2|p|^2 \le C\left(\int_0^T\int_M \theta^2|P_g p|^2 + \int_0^T\int_\omega \kappa^3\theta^2|p|^2\right). \tag{2.32}$$

This estimate holds for $s \ge aT + bT^2$, with a, b, C independent of T, s: take $a \ge \delta_0^{-1}$ and $4b \ge M_0$. The weight envelopes in (3.1) follow from $\psi < -1$, with $\vartheta = \theta$.

The argument in Section 3 now applies with $\mathrm{d}x$ replaced by $\mathrm{d}V_g$ and $c_f = \operatorname{Vol}_g(M)^{1/2}|f(0)|$. The duality, bridge and contraction arguments use only this Carleman estimate, finite volume and the unitary group of A. The same argument gives the shifted estimate (3.32), with a constant independent of the reference trajectory, since $z \mapsto f(u+z) - f(u)$ has Lipschitz constant L and vanishes at zero. Using the uniform threshold, we obtain (1.14) by choosing the parameter and subdividing time as in Section 3.3. This completes the proof of the proposition. □

Proof of Corollary 1.12. Write $\ell = \operatorname{Vol}_g(M)$. If $\partial M \ne \varnothing$, arc length identifies M isometrically with $[0, \ell]$. Indeed, in a coordinate with $g = a(x)^2\,\mathrm{d}x^2$, the change $s = \int_0^x a(r)\,\mathrm{d}r$ gives

$$\mathrm{d}V_g = \mathrm{d}s, \qquad \Delta_g = a^{-1}\partial_x(a^{-1}\partial_x) = \partial_s^2.$$

The change of variable preserves the pointwise nonlinearity, Dirichlet conditions, control supports, and every norm in the three theorems. Their interval statements therefore apply directly.

If $\partial M = \varnothing$, arc length identifies M with the circle $\mathbb{R}/\ell\mathbb{Z}$. Choose a cut point inside ω, so that $K = \overline{M \setminus \omega}$ is a compact subset of the coordinate interval $(0, \ell)$. We choose a smooth periodic function that agrees with $(s+1)^2$ near K, modifying it only near the cut inside ω. Its derivative

is nonzero and its second derivative equals 2 near K. Thus the intrinsic phase hypothesis holds, and Proposition 1.11 gives the Lipschitz conclusion and the shifted estimate (3.32) on the circle.

The remaining dispersive estimate follows from the periodic calculation in Lemma 4.2. For completeness, put $P_* = \ell^2/(2\pi)$ and write $v(s) = \sum_n c_n e^{2\pi i n s/\ell}$. For a Fourier polynomial, the resonance relations

$$n_1 + n_2 = n_3 + n_4, \qquad n_1^2 + n_2^2 = n_3^2 + n_4^2$$

imply $\{n_1, n_2\} = \{n_3, n_4\}$ and hence

$$\int_0^{P_*} \int_0^{\ell} |e^{-it\Delta_g} v|^4 \, \mathrm{d}s \, \mathrm{d}t = \ell P_* \left[2 \left(\sum_n |c_n|^2 \right)^2 - \sum_n |c_n|^4 \right] \le \frac{2P_*}{\ell} \|v\|_2^4.$$

Density and a finite covering by temporal periods give the free L^4 estimate uniformly for $0 < \tau \le T_*$, for each fixed $T_* > 0$. Minkowski's inequality gives the corresponding Duhamel estimate (4.7) for $L_t^1 L_x^2$ forcing.

The remaining arguments in Section 4 use only this estimate, finite spatial volume, unitarity of the free group, and the shifted Lipschitz control estimate. The truncation and mass estimates rely on pointwise and Hilbert-space arguments, and their volume factors remain $\ell^{1/2}$. The forced well-posedness proof and both control cascades therefore apply on the circle, with the same exponents, regularity, and uniqueness classes. The canonical logarithmic cost follows from the same scalar gauge transformation. □

3 Lipschitz exact controllability

First, we use the weighted duality to construct a linear control operator with zero endpoint states. To handle arbitrary endpoints, we join two uncontrolled trajectories and correct the resulting bridge by a contraction argument. Then, we subdivide the time interval to obtain the control cost in terms of the control time and the Lipschitz constant.

3.1 Weighted linear control. Fix $T > 0$, set $Q = (0, T) \times \Omega$ and $Q_\omega = (0, T) \times \omega$, and write $H = L^2(\Omega; \mathbb{C})$. The inner product $\langle \cdot, \cdot \rangle$ on H is linear in its first argument. We use the Dirichlet realization $A = -\Delta_D$ and its unitary group $U(t) = e^{itA}$, so that $P = i\partial_t + A = i\partial_t - \Delta$. We formulate the geometric assumptions using a common weight for the state and source, together with a separate observation weight. This formulation includes the folded interval weights of Proposition 2.11; the observation weight determines the control weight in the dual construction.

Set

$$\kappa(t) = \frac{s}{t(T-t)}, \qquad s \ge s_0 > 0.$$

Suppose that positive measurable weights $\theta = \theta_s$ and $\vartheta = \vartheta_s$ on Q satisfy

$$c_* e^{-\beta\kappa} \le \theta \le e^{-\alpha\kappa}, \qquad 0 < \vartheta \le e^{-\alpha\kappa}, \tag{3.1}$$

where the inequalities hold almost everywhere and $c_*, \alpha, \beta > 0$ are independent of s. No lower envelope for ϑ or comparison between the two weights is needed. Suppose, in addition, that

$$\int_Q \kappa^3 \theta^2 |q|^2 \, \mathrm{d}x \, \mathrm{d}t \le C_{\mathrm{Car}} \left(\int_Q \theta^2 |Pq|^2 \, \mathrm{d}x \, \mathrm{d}t + \int_{Q_\omega} \kappa^3 \vartheta^2 |q|^2 \, \mathrm{d}x \, \mathrm{d}t \right) \tag{3.2}$$

for smooth Dirichlet test functions q, with a constant $C_{\mathrm{Car}} > 0$ independent of $s \ge s_0$. The test functions have unrestricted values at $t = 0$ and $t = T$. The freedom to choose these

values independently enforces both endpoint conditions on the controlled state. Proposition 2.4, Proposition 2.8, and Proposition 2.11 ensure these hypotheses in the geometric configurations considered here. The argument below uses only (3.1) and (3.2).

For this fixed value of s, define the weighted Hilbert spaces

$$\begin{aligned}\mathcal{S}_s &= L^2(Q;\kappa^{-3}\theta^{-2}\,\mathrm{d}x\,\mathrm{d}t),\\ \mathcal{Y}_s &= L^2(Q;\theta^{-2}\,\mathrm{d}x\,\mathrm{d}t),\\ \mathcal{U}_s &= L^2(Q_\omega;\kappa^{-3}\vartheta^{-2}\,\mathrm{d}x\,\mathrm{d}t).\end{aligned}$$

These spaces consist of complex-valued functions. We equip intersections of Banach spaces with the sum norm. We also set

$$m_s = \inf_{0<t<T}\kappa(t) = \frac{4s}{T^2}, \qquad B_s = \operatorname*{ess\,sup}_{(t,x)\in Q}\kappa(t)^3\max\{\theta(t,x)^2,\vartheta(t,x)^2\} < \infty.$$

The finiteness of B_s follows from (3.1), since $\sup_{a>0} a^3e^{-2\alpha a} < \infty$. Consequently,

$$\|F\|_{L^2(Q)} \le B_s^{1/2}\|F\|_{\mathcal{S}_s}, \qquad \|h\|_{L^2(Q_\omega)} \le B_s^{1/2}\|h\|_{\mathcal{U}_s}, \qquad \|v\|_{L^2(Q)} \le \|v\|_{\mathcal{Y}_s}. \tag{3.3}$$

In formulas on Q, a control on Q_ω is extended by zero and denoted by $\mathbf{1}_\omega h$.

Proposition 3.1. *Assume (3.1) and (3.2). For every $s \ge s_0$ there is a bounded linear map*

$$\Lambda_s : \mathcal{S}_s \longrightarrow (\mathcal{Y}_s \cap C([0,T];H)) \times \mathcal{U}_s, \qquad F \longmapsto (v_F, h_F),$$

such that

$$Pv_F = F + \mathbf{1}_\omega h_F, \qquad v_F(0) = v_F(T) = 0.$$

The equation and the homogeneous Dirichlet boundary condition are understood in the mild sense associated with A. Moreover,

$$\|v_F\|_{\mathcal{Y}_s}^2 + \|h_F\|_{\mathcal{U}_s}^2 \le C_{\mathrm{Car}}\|F\|_{\mathcal{S}_s}^2, \tag{3.4}$$

and

$$\|v_F\|_{C([0,T];H)} \le \sqrt{T}\,B_s^{1/2}(1+\sqrt{C_{\mathrm{Car}}})\|F\|_{\mathcal{S}_s}. \tag{3.5}$$

Proof. Let

$$\mathcal{D} = \{q \in C^\infty([0,T]\times\overline{\Omega};\mathbb{C}) : q = 0 \text{ on } [0,T]\times\partial\Omega\}.$$

On this space, define the Hermitian form

$$a_s(p,q) = \int_Q \theta^2 Pp\,\overline{Pq}\,\mathrm{d}x\,\mathrm{d}t + \int_{Q_\omega}\kappa^3\vartheta^2 p\bar{q}\,\mathrm{d}x\,\mathrm{d}t. \tag{3.6}$$

Estimate (3.2) shows that a_s is positive definite. Let $\mathcal{H}_s$ be the completion of $\mathcal{D}$ with respect to the norm $a_s(q,q)^{1/2}$.

We first realize this completion as a space of functions. Let (p_j) be an a_s-Cauchy sequence. The sequences $\kappa^{3/2}\theta p_j$ and θPp_j converge in $L^2(Q)$; for the first sequence, this follows from (3.2). The sequence $\kappa^{3/2}\vartheta p_j$ also converges in $L^2(Q_\omega)$ by the definition of a_s. On each compact subset of Q, θ and $\kappa^{3/2}\theta$ are bounded above and below by positive constants in view of (3.1). Hence $p_j \to p$ in $L^2_{\mathrm{loc}}(Q)$, and the local L^2 limit of Pp_j agrees with Pp in the sense of distributions. The local boundedness of $\kappa^{3/2}\vartheta$ identifies the local limit of the observation sequence as $\kappa^{3/2}\vartheta p$. This agrees with its global $L^2(Q_\omega)$ limit. The same argument shows that these identifications do not

depend on the approximating sequence. Therefore, θPp, $\kappa^{3/2}\theta p$, and $\kappa^{3/2}\vartheta p|_{Q_\omega}$ are well-defined L^2 functions, and both (3.6) and (3.2) extend to $\mathcal{H}_s$. This realization is injective. Indeed, if the local representative p vanishes, then $Pp = 0$ in the sense of distributions. Both weighted L^2 limits defining a_s therefore vanish, so the corresponding completed element has zero $\mathcal{H}_s$ norm. We have thus obtained a continuous embedding of $\mathcal{H}_s$ into $L^2_{\mathrm{loc}}(Q)$.

For $F \in \mathcal{S}_s$, the conjugate-linear functional

$$\ell_F(q) = \int_Q F\bar{q}\ \mathrm{d}x\ \mathrm{d}t$$

is continuous on $\mathcal{H}_s$, because

$$|\ell_F(q)| \le \|F\|_{\mathcal{S}_s}\left(\int_Q \kappa^3\theta^2|q|^2\ \mathrm{d}x\ \mathrm{d}t\right)^{1/2} \le \sqrt{C_{\mathrm{Car}}}\|F\|_{\mathcal{S}_s}a_s(q,q)^{1/2}.$$

By the Riesz representation theorem, there exists a unique $p_F \in \mathcal{H}_s$ such that

$$a_s(p_F, q) = \ell_F(q) \qquad (q \in \mathcal{H}_s). \tag{3.7}$$

Define

$$v_F = \theta^2 P p_F, \qquad h_F = -\kappa^3\vartheta^2 p_F|_{Q_\omega}. \tag{3.8}$$

Taking $q = p_F$ in (3.7) gives

$$\|v_F\|^2_{\mathcal{Y}_s} + \|h_F\|^2_{\mathcal{U}_s} = a_s(p_F, p_F) \le \sqrt{C_{\mathrm{Car}}}\|F\|_{\mathcal{S}_s}a_s(p_F, p_F)^{1/2},$$

which proves (3.4). The construction is linear in F.

It remains to show that the state constructed by duality is a mild solution with zero initial and final states. Set $G = F + \mathbf{1}_\omega h_F$. It follows from (3.3) that $G \in L^2(0,T;H)$ and $v_F \in L^2(0,T;H)$. Combining (3.7) with (3.8), we obtain

$$\int_Q v_F\overline{Pq}\ \mathrm{d}x\ \mathrm{d}t = \int_Q G\bar{q}\ \mathrm{d}x\ \mathrm{d}t \qquad (q \in \mathcal{D}). \tag{3.9}$$

For smooth Dirichlet functions v, q, the underlying Green identity is

$$\int_Q v\overline{Pq}\ \mathrm{d}x\ \mathrm{d}t = \int_Q (Pv)\bar{q}\ \mathrm{d}x\ \mathrm{d}t - i[\langle v(t), q(t)\rangle]_0^T.$$

Since the temporal traces of v_F are not yet available, we recover them from (3.9), coefficient by coefficient in a Dirichlet eigenbasis. Let $(e_k)_{k\ge1}$ be an orthonormal Dirichlet eigenbasis of H, with $Ae_k = \lambda_k e_k$. Smoothness of the domain gives $e_k \in C^\infty(\overline{\Omega})$, so $q(t,x) = \zeta(t)e_k(x)$ is admissible for every $\zeta \in C^\infty([0,T])$. Write $v_k(t) = \langle v_F(t), e_k\rangle$ and $G_k(t) = \langle G(t), e_k\rangle$. Then

$$\int_0^T v_k(t)(-i\overline{\zeta'(t)} + \lambda_k\overline{\zeta(t)})\ \mathrm{d}t = \int_0^T G_k(t)\overline{\zeta(t)}\ \mathrm{d}t. \tag{3.10}$$

Taking $\zeta \in C_c^\infty(0,T)$ first, we obtain

$$iv_k' + \lambda_k v_k = G_k$$

in distributions. Since both v_k and G_k belong to $L^2(0,T)$, we have $v_k \in H^1(0,T)$. We can therefore integrate by parts in (3.10) for arbitrary $\zeta \in C^\infty([0,T])$. The differential equation cancels the interior terms, leaving

$$-i[v_k(t)\overline{\zeta(t)}]_{t=0}^{t=T} = 0.$$

As the two endpoint values of ζ can be chosen independently,

$$v_k(0) = v_k(T) = 0 \qquad (k \geq 1). \tag{3.11}$$

Define the Bochner integral

$$w(t) = -i \int_0^t U(t-\tau)G(\tau)\, \mathrm{d}\tau. \tag{3.12}$$

Strong continuity and unitarity of U imply that $w \in C([0,T]; H)$ and

$$\|w\|_{C([0,T];H)} \leq \int_0^T \|G(t)\|_H \, \mathrm{d}t \leq \sqrt{T}\|G\|_{L^2(Q)}.$$

For each k, the coefficient $\langle w(t), e_k\rangle$ and v_k solve the same scalar initial-value problem with zero initial data. Scalar uniqueness and completeness of the eigenbasis give $v_F = w$ almost everywhere on Q. We henceforth use this continuous representative of v_F. Its terminal coefficients vanish by (3.11); completeness of the eigenbasis then gives $v_F(T) = 0$. Formula (3.12) also yields the Dirichlet mild formulation. Finally, (3.3) and (3.4) give

$$\|G\|_{L^2(Q)} \leq B_s^{1/2}(1 + \sqrt{C_{\mathrm{Car}}})\|F\|_{\mathcal{S}_s},$$

which proves (3.5). □

Remark 3.2. For an L^2 state, the Dirichlet boundary condition is encoded by the group $U(t)$ and need not be understood as a classical spatial trace. The temporal endpoint conditions follow from the scalar H^1 regularity and the unrestricted endpoint values of the test functions in (3.10).

3.2 Arbitrary endpoints. Assume (1.2), with Lipschitz constant $L \geq 0$. As above, f also denotes the induced Nemytskii map on H. Since Ω has finite measure,

$$\|f(u) - f(v)\|_H \leq L\|u - v\|_H, \qquad \|f(u)\|_H \leq L\|u\|_H + c_f, \qquad c_f = |\Omega|^{1/2}|f(0)|. \tag{3.13}$$

In particular, no condition $f(0) = 0$ is imposed.

Lemma 3.3. *Assume (1.2), and fix $T > 0$. For every $y_0, y_1 \in H$, there exist $u^0, u^1 \in C([0,T]; H)$ satisfying $Pu^j = f(u^j)$ in the Dirichlet mild sense and*

$$u^0(0) = y_0, \qquad u^1(T) = y_1.$$

There is a bridge $r \in C([0,T]; H)$ with $r(0) = y_0$, $r(T) = y_1$ such that

$$R = f(r) - Pr \in \mathcal{S}_s \qquad \text{for every fixed } s \geq s_0. \tag{3.14}$$

More precisely, $R \in C([0,T]; H)$ and $R(t) = 0$ for $t \in [0, T/3] \cup [2T/3, T]$.

Proof. For data $z \in H$ at any time $t_0 \in [0,T]$, the mild equation is

$$u(t) = U(t - t_0)z - i \int_{t_0}^t U(t-\tau)f(u(\tau))\, \mathrm{d}\tau,$$

where the integral for $t < t_0$ is interpreted with the usual orientation. By (3.13), the right-hand side is a contraction on sufficiently short intervals in $C(H)$. We iterate this construction in either time direction. The linear growth bound prevents blow-up on $[0,T]$. More precisely, set

$$\Phi_L(a) = \begin{cases} (e^{La} - 1)/L, & L > 0, \\ a, & L = 0, \end{cases} \qquad a \geq 0,$$

and apply the integral estimate and Gronwall's inequality to obtain

$$\|u(t)\|_H \le e^{L|t-t_0|}\|z\|_H + c_f\Phi_L(|t-t_0|). \tag{3.15}$$

This proves existence of both the forward trajectory u^0 and the backward trajectory u^1.

Choose a real-valued function $\eta \in C^\infty([0,T])$ such that

$$0 \le \eta \le 1, \qquad \eta = 1 \text{ on } [0,T/3], \qquad \eta = 0 \text{ on } [2T/3,T], \qquad \|\eta'\|_{L^\infty} \le c_\eta/T, \tag{3.16}$$

where c_η is an absolute constant determined by a fixed cutoff. Set

$$r = \eta u^0 + (1-\eta)u^1.$$

By the product rule, interpreted in the mild equations or their weak formulation against $D(A)$, we obtain

$$Pr = \eta f(u^0) + (1-\eta)f(u^1) + i\eta'(u^0-u^1).$$

Thus the residual in (3.14) is the $C([0,T];H)$ function

$$R = f(r) - \eta f(u^0) - (1-\eta)f(u^1) - i\eta'(u^0-u^1). \tag{3.17}$$

It vanishes for $t \le T/3$ and for $t \ge 2T/3$. On the intervening closed time interval, the weight $\kappa^{-3}\theta^{-2}$ is bounded for each fixed s. It follows that $R \in \mathcal{S}_s$. □

Proof of exact controllability in Theorem 1.4. Under the geometric assumption of the theorem, the corresponding Carleman estimate gives weights satisfying (3.1) and (3.2). In particular, for an arbitrary nonempty open control set on an interval, take $\theta = W_s^{1/2}$ and $\vartheta = V_s^{1/2}$ from Proposition 2.11; then $c_* = 1/2$. Fix $y_0, y_1 \in H$ and construct r and R as in Lemma 3.3. For $F \in \mathcal{S}_s$, let $(v_F, h_F) = \Lambda_s F$ be the pair given by Proposition 3.1. To correct the bridge, define

$$\mathcal{N}_s(F) = R + f(r+v_F) - f(r). \tag{3.18}$$

The pointwise Lipschitz estimate gives

$$\begin{aligned}\|f(r+v_F)-f(r)\|_{\mathcal{S}_s}^2 &\le L^2 \int_Q \kappa^{-3}\theta^{-2}|v_F|^2 \,\mathrm{d}x\,\mathrm{d}t \\ &\le L^2 m_s^{-3}\|v_F\|_{\mathcal{Y}_s}^2.\end{aligned}$$

Since $R \in \mathcal{S}_s$, it follows that $\mathcal{N}_s$ maps $\mathcal{S}_s$ into itself. Applying the same estimate to the difference of two sources and using the linearity of Λ_s together with (3.4), we obtain

$$\|\mathcal{N}_s(F) - \mathcal{N}_s(\widetilde{F})\|_{\mathcal{S}_s} \le q_s\|F-\widetilde{F}\|_{\mathcal{S}_s}, \qquad q_s = L\sqrt{C_{\mathrm{Car}}}\, m_s^{-3/2}. \tag{3.19}$$

Since C_{Car} is independent of s, we can choose $s \ge s_0$ such that $q_s < 1$. This choice imposes no smallness condition on L or the endpoint states.

By the Banach fixed-point theorem, there exists a unique fixed point $F_* \in \mathcal{S}_s$ for this choice of bridge and control operator, satisfying

$$F_* = R + f(r+v_{F_*}) - f(r).$$

Since $\mathcal{N}_s(0) = R$, the fixed point satisfies

$$\|F_*\|_{\mathcal{S}_s} \le \frac{\|R\|_{\mathcal{S}_s}}{1-q_s}. \tag{3.20}$$

Define

$$y = r + v_{F_*}, \qquad h = h_{F_*}.$$

The bridge and the endpoint conditions for the linear correction give $y \in C([0,T];H)$, $y(0) = y_0$, and $y(T) = y_1$. Moreover,

$$\begin{aligned} Py &= Pr + F_* + \mathbf{1}_\omega h \\ &= Pr + R + f(r + v_{F_*}) - f(r) + \mathbf{1}_\omega h \\ &= f(y) + \mathbf{1}_\omega h. \end{aligned}$$

The right-hand side belongs to $L^2(0,T;H)$ and hence to $L^1(0,T;H)$. Adding the Dirichlet mild formulations for r and v_{F_*} therefore yields

$$y(t) = U(t)y_0 - i\int_0^t U(t-\tau)(f(y(\tau)) + \mathbf{1}_\omega h(\tau))\,\mathrm{d}\tau.$$

Finally, $h \in L^2(Q_\omega)$ by (3.3). For this fixed control, uniqueness of the mild solution follows from (3.13) and Gronwall's inequality. This proves the asserted exact controllability for arbitrary L^2 endpoints. The uniform bound (1.5) is proved in Section 3.3. □

Proposition 3.4. *Under the assumptions of Theorem 1.4, let r, R, Λ_s, and $\mathcal{N}_s$ be the bridge, residual, control operator, and nonlinear source map constructed above, respectively. Fix $s \geq s_0$ with $q_s = L\sqrt{C_{\mathrm{Car}}}m_s^{-3/2} < 1$. Let F_* be the unique fixed point of $\mathcal{N}_s$, and set $y = r + v_{F_*}$ and $h = h_{F_*}$. Define*

$$F^{(0)} = 0, \qquad F^{(n+1)} = \mathcal{N}_s(F^{(n)}), \qquad (v^{(n)}, h^{(n)}) = \Lambda_s F^{(n)}, \qquad y^{(n)} = r + v^{(n)}.$$

Then $F^{(n)} \to F_*$ in $\mathcal{S}_s$, $v^{(n)} \to v_{F_*}$ in $\mathcal{Y}_s \cap C([0,T];H)$, and $h^{(n)} \to h$ in $\mathcal{U}_s$ and in $L^2(Q_\omega)$. More precisely, for every $n \geq 0$, with $q_s^0 = 1$,

$$\|F^{(n)} - F_*\|_{\mathcal{S}_s} \leq \frac{q_s^n}{1-q_s}\|R\|_{\mathcal{S}_s}, \tag{3.21}$$

and

$$\left(\|v^{(n)} - v_{F_*}\|_{\mathcal{Y}_s}^2 + \|h^{(n)} - h\|_{\mathcal{U}_s}^2\right)^{1/2} \leq \frac{\sqrt{C_{\mathrm{Car}}}q_s^n}{1-q_s}\|R\|_{\mathcal{S}_s}. \tag{3.22}$$

The convergence in $C([0,T];H)$ satisfies the estimate

$$\|y^{(n)} - y\|_{C([0,T];H)} \leq \sqrt{T}\,B_s^{1/2}(1 + \sqrt{C_{\mathrm{Car}}})\frac{q_s^n}{1-q_s}\|R\|_{\mathcal{S}_s}. \tag{3.23}$$

Every iterate satisfies the endpoint conditions $y^{(n)}(0) = y_0$ and $y^{(n)}(T) = y_1$ exactly, and its nonlinear equation has the residual

$$Py^{(n)} - f(y^{(n)}) - \mathbf{1}_\omega h^{(n)} = F^{(n)} - F^{(n+1)}, \qquad \|F^{(n)} - F^{(n+1)}\|_{\mathcal{S}_s} \leq q_s^n\|R\|_{\mathcal{S}_s}. \tag{3.24}$$

Proof. Because $F^{(1)} = R$, estimate (3.19) gives

$$\|F^{(n+1)} - F^{(n)}\|_{\mathcal{S}_s} \leq q_s^n\|R\|_{\mathcal{S}_s}.$$

Summing this geometric bound from n to infinity yields (3.21). By the linearity of Λ_s and (3.4), we obtain (3.22), while (3.5) yields (3.23). The control also converges in the unweighted space by (3.3). Each iterate has the required endpoint values by Proposition 3.1. Substituting (3.14) and (3.18) gives the identity in (3.24). □

Proposition 3.5. *Under the assumptions of Theorem 1.4, let r and R be the bridge and residual in its proof. For each $s \geq s_0$ with $q_s < 1$, the control constructed from these data satisfies*

$$\|h\|_{L^2(Q_\omega)} \leq \frac{\sqrt{C_{\mathrm{Car}} B_s}}{1 - q_s} \|R\|_{\mathcal{S}_s}. \tag{3.25}$$

There is a choice of s, depending on L and on the fixed geometric and temporal parameters, for which

$$\begin{aligned} \|h\|_{L^2(Q_\omega)} \leq C_T(1+L)\exp\big(LT + C_T(1+L^{2/3})\big) \\ \times \left(\|y_0\|_H + \|y_1\|_H + T|\Omega|^{1/2}|f(0)|\right). \end{aligned} \tag{3.26}$$

Here C_T can depend on Ω, ω, T and the fixed constants $s_0, C_{\mathrm{Car}}, c_*, \alpha, \beta$ in (3.1)–(3.2). It is independent of f, L, y_0 and y_1.

Proof. Estimate (3.25) follows by combining (3.3), (3.4), and (3.20).

To obtain the explicit bound, use α, β, c_* from (3.1) and set

$$m_0 = \max\left\{\frac{4s_0}{T^2}, 1, \frac{3}{2\alpha}\right\}.$$

Choose

$$m = \max\{m_0, (2L\sqrt{C_{\mathrm{Car}}})^{2/3}\}, \qquad s = \frac{T^2 m}{4}. \tag{3.27}$$

Then $s \geq s_0$, $m_s = m$, and $q_s \leq 1/2$. Moreover, $m \leq C_T(1 + L^{2/3})$. Since $a \mapsto a^3 e^{-2\alpha a}$ is decreasing for $a \geq 3/(2\alpha)$,

$$B_s^{1/2} \leq m^{3/2} e^{-\alpha m}. \tag{3.28}$$

On the support of R, the choice of cutoff gives

$$m \leq \kappa(t) \leq \frac{9}{8} m.$$

It follows that

$$\|R\|_{\mathcal{S}_s} \leq c_*^{-1} m^{-3/2} e^{9\beta m/8} \|R\|_{L^2(Q)}. \tag{3.29}$$

We next estimate the unweighted residual. Set $d = \|y_0\|_H + \|y_1\|_H$. By (3.15), we have, for $0 \leq t \leq T$,

$$\begin{aligned} \|u^0(t) - u^1(t)\|_H &\leq e^{Lt}\|y_0\|_H + e^{L(T-t)}\|y_1\|_H + c_f(\Phi_L(t) + \Phi_L(T-t)) \\ &\leq e^{LT} d + c_f \Phi_L(T) \leq e^{LT}(d + T c_f). \end{aligned}$$

For $L > 0$, the middle inequality follows from $(e^{Lt} - 1)(e^{L(T-t)} - 1) \geq 0$; for $L = 0$, it is immediate. For the nonlinear part of the residual, the pointwise Lipschitz condition gives

$$\begin{aligned} &|f(\eta u^0 + (1-\eta)u^1) - \eta f(u^0) - (1-\eta) f(u^1)| \\ &\quad \leq \eta L(1-\eta)|u^0 - u^1| + (1-\eta)L\eta|u^0 - u^1| \leq \frac{L}{2}|u^0 - u^1|. \end{aligned}$$

Using (3.17), the support of R, and (3.16), we obtain

$$\|R\|_{L^2(Q)} \leq \sqrt{T/3}\left(\frac{L}{2} + \frac{c_\eta}{T}\right) e^{LT}(d + T c_f) \leq C_T(1+L) e^{LT}(d + T c_f). \tag{3.30}$$

Combining (3.25)– (3.30) and using $(1 - q_s)^{-1} \leq 2$, we cancel the factors $m^{3/2}$ and $m^{-3/2}$ and obtain

$$\|h\|_{L^2(Q_\omega)} \leq C_T(1+L) e^{LT} e^{(9\beta/8 - \alpha)m}(d + T c_f).$$

Using the bound on m following (3.27) and increasing C_T if necessary, we obtain (3.26). □

Remark 3.6. The term involving $f(0)$ in (3.26) accounts for the forcing present even at the zero state. It cannot in general be omitted from a bound valid for all globally Lipschitz nonlinearities and all endpoint data. For example, if $f \equiv c \neq 0$ and $y_0 = y_1 = 0$, the uncontrolled final state is $-ic \int_0^T U(T-t)\mathbf{1}_\Omega \, \mathrm{d}t$, which is nonzero at nonresonant times.

The nonlinear endpoint trajectories in Lemma 3.3 ensure that the residual vanishes near the endpoints. For a general f, replacing these trajectories by solutions of $Pu = f(0)$ leaves the residual $f(r) - f(0)$ near the endpoints. This residual need not belong to $\mathcal{S}_s$ because of the exponentially growing weight. If f is real-affine, its affine flow provides the required endpoint trajectories. For example, when $f(z) = az + b\bar{z} + c$, the real cutoff satisfies $f(r) = \eta f(u^0) + (1-\eta) f(u^1)$, so the residual reduces exactly to $R = -i\eta'(u^0 - u^1)$.

Remark 3.7. The Lipschitz contraction argument alone does not apply to a general locally Lipschitz or superlinear nonlinearity. For example, if $g(z) = |z|^{p-1}z$ with $p > 1$, the available difference estimate is

$$|g(z) - g(w)| \leq C_p(|z| + |w|)^{p-1}|z - w|.$$

The factor involving the states is not controlled by the $\mathcal{Y}_s$ norm. A weighted L^2 estimate does not imply a pointwise bound, and g does not map every $L^2(\Omega)$ state into $L^2(\Omega)$. Thus a large Carleman parameter alone does not make this map contractive on the source space used here. Truncation gives a controlled solution of the truncated equation, but passing to the original equation requires an additional bound uniform in the truncation level.

For the logarithmic class in Section 4, we use the truncation and stability argument given below.

3.3 Uniform control costs.

Proof of the cost bound in Theorem 1.4. We first derive a control estimate on a single interval of length $0 < \tau \leq 1$. We then concatenate the resulting controls, choosing the interval length to limit the growth of the uncontrolled trajectories. Put $c_f = |\Omega|^{1/2}|f(0)|$, $a_j = \tau c_f$, and $d_j = \|z_0\|_2 + \|z_1\|_2$ for the two endpoints. The bridge in Lemma 3.3, together with (3.15) and (3.30), gives

$$\|r\|_{CL^2} \leq e^{L\tau}(d_j + a_j), \qquad \|R\|_{L^2} \leq C\sqrt{\tau}(L + \tau^{-1})e^{L\tau}(d_j + a_j).$$

For the linear map in Proposition 3.1, define

$$\mathcal{N}_s(F) = R + f(r + v_F) - f(r).$$

Writing $m = 4s/\tau^2$, the weighted embedding and (3.4) give

$$\|\mathcal{N}_s(F) - \mathcal{N}_s(G)\|_{\mathcal{S}_s} \leq L\sqrt{C_{\mathrm{Car}}}\, m^{-3/2}\|F - G\|_{\mathcal{S}_s}.$$

The nonlinear difference belongs to the weighted source space, and $\mathcal{N}_s(0) = R$. When the factor above is less than one, this map has a unique fixed point, which gives the controlled pair $y = r + v_F$, $h = h_F$. The growth bound (3.13) ensures that the controlled state satisfies the mild equation (1.4). Using the uniform Carleman threshold, choose

$$m = \max\left\{\frac{4a}{\tau} + 4b,\ 1,\ \frac{3}{2\alpha},\ (2L\sqrt{C_{\mathrm{Car}}})^{2/3}\right\}, \qquad s = \frac{\tau^2 m}{4}.$$

Then the contraction factor is at most $1/2$ and $m \leq C(\tau^{-1} + L^{2/3})$. Combining (3.28) with (3.29), we cancel the factors $m^{3/2}$ and $m^{-3/2}$. Using the resulting control bound and (3.5), we obtain

$$\|h\|_2 \leq C\sqrt{\tau}(L + \tau^{-1})e^{Cm + L\tau}(d_j + a_j),$$
$$\|v\|_{CL^2} \leq C\tau(L + \tau^{-1})e^{Cm + L\tau}(d_j + a_j).$$

Since $y = r + v$, we obtain, after absorbing these polynomial factors,

$$\|h\|_2 + \|y\|_{CL^2} \le Ce^{C(\tau^{-1}+L^{2/3})+L\tau}(d_j + a_j). \tag{3.31}$$

We now let $T > 0$ be arbitrary. To remove the factor e^{LT}, set

$$J = \max\{1, \lceil T(1+L)^{1/3}\rceil\}, \qquad \tau = T/J.$$

Then $0 < \tau \le 1$, $\tau^{-1} \le T^{-1} + (1+L)^{1/3}$ and $L\tau \le (1+L)^{2/3}$. Hence the prefactor in each segment estimate is bounded by the same constant $A = Ce^{C(T^{-1}+L^{2/3})}$. If $J = 1$, apply (3.31) directly. If $J \ge 2$, join y_0 to zero on the first interval, zero to zero on all intervening intervals, and zero to y_1 on the last interval. The single-interval construction supplies the controls on the intervening intervals even when $f(0) \ne 0$. The corresponding trajectories have zero endpoint states, although they need not vanish between the endpoints.

On each segment, the forcing belongs to $L^1_t H$ and the control belongs to $L^2_t L^2(\omega)$. The states agree at every junction. Thus, by repeated use of the group identity $U(t-r)U(r-q) = U(t-q)$, we obtain the mild equation on the whole interval, with no distributional jump term. The global control norm is the Euclidean norm of the segment control norms. Since $\sum_j a_j = Tc_f$ and the first and last d_j are the only ones that can be nonzero, the triangle inequality yields

$$\|h\|_2 \le A\Big(\sum_j (d_j + a_j)^2\Big)^{1/2} \le A(\|y_0\|_2 + \|y_1\|_2 + Tc_f).$$

The supremum norm of the state is bounded by $A \max_j(d_j + a_j)$, which satisfies the same final bound. Increasing C yields (1.5) for every $T > 0$. For a prescribed control, the global Lipschitz bound and Gronwall's inequality give uniqueness of the mild solution in $C([0,T]; H)$. □

Remark 3.8. No conservation law is required in Theorem 1.4. If f is mass preserving, its uncontrolled trajectories have constant norm. The single-interval estimate then avoids $e^{L\tau}$. For $T \le 1$, it gives the uniform cost directly. For $T > 1$, we follow the uncontrolled flow until $T - 1$ and apply the single-interval estimate on the last unit interval. The norm is unchanged during the uncontrolled evolution, so the same uniform bound holds without concatenating several controlled intervals. For continuous scalar f, $\mathrm{Im}(f(z)\bar{z}) = 0$ for every z also implies $f(0) = 0$, as seen by testing $z = r$ and $z = ir$ and letting $r \to 0$.

For the logarithmic truncations, we need a control estimate for a nonlinear difference along a prescribed reference function.

Lemma 3.9. *Assume one of the geometric conditions in Theorem 1.4, and let $f : \mathbb{C} \to \mathbb{C}$ satisfy Assumption 1.1 with Lipschitz constant L. Fix $T > 0$ and a measurable function $u : Q \to \mathbb{C}$ that is finite almost everywhere. For every $z_0, z_1 \in H$, there is a control $h \in L^2(Q_\omega; \mathbb{C})$ such that the Dirichlet problem*

$$Pz = f(u+z) - f(u) + \mathbf{1}_\omega h, \qquad z(0) = z_0,$$

has a unique mild solution $z \in C([0,T]; H)$ satisfying $z(T) = z_1$. For every $T > 0$, the control and solution can be chosen so that

$$\|h\|_{L^2(Q_\omega)} + \|z\|_{C([0,T];H)} \le Ce^{C(T^{-1}+L^{2/3})}(\|z_0\|_H + \|z_1\|_H). \tag{3.32}$$

Here C depends only on the fixed spatial geometry, the control region and, where applicable, the phase data. In particular, it is independent of u, $f(0)$, T, L and the endpoint states.

Proof. The pointwise inequalities

$$|f(u+v)-f(u+w)| \le L|v-w|,$$
$$|f(u+v)-f(u)| \le L|v|$$

show that the nonlinear difference defines a strongly measurable H-valued forcing along every continuous H-valued trajectory. This forcing is globally Lipschitz in the trajectory and vanishes at zero. The Duhamel contraction on short time intervals can therefore be iterated to give unique forward and backward uncontrolled trajectories. No integrability of u is needed.

As in the preceding proof, we first work on an interval of length $0 < \tau \le 1$ and translate its left endpoint to zero. For the prescribed endpoints, set $d = \|z_0\|_H + \|z_1\|_H$. Let v^0 and v^1 be the uncontrolled trajectories starting from z_0 and ending at z_1, respectively. By Gronwall's inequality,

$$\|v^0\|_{CH} + \|v^1\|_{CH} \le e^{L\tau} d.$$

Choose the cutoff η as in (3.16), with T replaced by τ, and set $r = \eta v^0 + (1-\eta)v^1$. Its residual is

$$\begin{aligned} R = {} & f(u+r) - f(u) - \eta[f(u+v^0) - f(u)] \\ & - (1-\eta)[f(u+v^1) - f(u)] - i\eta'(v^0 - v^1). \end{aligned}$$

The mild equations imply $Pr = f(u+r) - f(u) - R$. Moreover, R vanishes outside $[\tau/3, 2\tau/3]$ and

$$\|r\|_{CH} \le e^{L\tau} d, \qquad \|R\|_{L^2((0,\tau)\times\Omega)} \le C\sqrt{\tau}(L + \tau^{-1}) e^{L\tau} d.$$

Indeed, the nonlinear part of R is bounded pointwise by $2L\eta(1-\eta)|v^0 - v^1|$. The support property also gives $R \in \mathcal{S}_s$ for every admissible Carleman parameter.

For $(v_F, h_F) = \Lambda_s F$, define the nonlinear correction map

$$\mathcal{N}_s(F) = R + f(u + r + v_F) - f(u + r).$$

By (3.3) and (3.4), its Lipschitz constant on $\mathcal{S}_s$ is at most $L\sqrt{C_{\mathrm{Car}}} m^{-3/2}$, where $m = 4s/\tau^2$. Choose s exactly as in the proof of (3.31), so that this constant is at most $1/2$ and $m \le C(\tau^{-1} + L^{2/3})$. The fixed point gives $z = r + v_F$ with the required endpoints. Using the residual bound above together with (3.28), (3.29) and (3.5), we obtain

$$\|h\|_2 + \|z\|_{CH} \le C e^{C(\tau^{-1} + L^{2/3}) + L\tau} d.$$

The nonlinear terms are estimated using only the two displayed Lipschitz bounds. Thus the estimates are independent of u and $f(0)$.

To pass to arbitrary $T > 0$, set $J = \max\{1, \lceil T(1+L)^{1/3} \rceil\}$ and $\tau = T/J$. Then $0 < \tau \le 1$, $\tau^{-1} \le T^{-1} + (1+L)^{1/3}$ and $L\tau \le (1+L)^{2/3}$. If $J \ge 2$, join z_0 to zero on the first interval and zero to z_1 on the last, taking both the solution and the control to be zero on the intervening intervals. The concatenated trajectories satisfy the mild equation on $[0, T]$. Taking the Euclidean norm of the segment control norms and the supremum of the state norms gives (3.32). For $J = 1$, the single-interval estimate gives the same conclusion. Uniqueness for the prescribed control follows from the Lipschitz difference bound and Gronwall's inequality. □

4 Logarithmic nonlinearities

Throughout this section, let $\Omega = (0, \ell)$ and let ω be a nonempty open subset of Ω. We first assume that Assumption 1.6 holds and write

$$f(z) = F(z) + \mathfrak{g}(z), \qquad F(z) = z\phi(|z|), \quad F(0) = 0, \qquad \mathrm{Lip}(\mathfrak{g}) \le L_g.$$

The exponent $0 \le \gamma < 1/2$ controls the derivative of the radial coefficient with respect to logarithmic amplitude. The critical case $\gamma = 1/2$ is treated separately in Subsection 4.3. Neither $f(0) = 0$ nor a conservation law is assumed for the full nonlinearity. To reach an arbitrary terminal state, we first construct an uncontrolled reference trajectory ending at that state. We then successively control the difference from this reference. At each step, we compare the true trajectory with the truncated controlled trajectory through a common truncation.

4.1 Truncation and the forced equation. For the Cauchy theory of the logarithmic Schrödinger equation, see [9, 10, 13]. The truncation and well-posedness lemmas in this subsection remain valid for $0 \le \gamma < 1$ under the same derivative bound. In particular, they apply at $\gamma = 1/2$. The restriction $\gamma < 1/2$ is needed for the control construction in the next subsection.

The stability estimate below extends the imaginary-part cancellation for the canonical logarithmic nonlinearity; see, for example, [8, Lemma 2.1 and Section 2.2]. We keep the dependence on the truncation level explicit for its later use in the control construction.

Lemma 4.1. *For $K \ge 1$, we define*

$$\rho_K(r) = \min\{e^K, \max\{e^{-K}, r\}\}, \qquad \phi_K(r) = \phi(\rho_K(r)), \qquad f_K(z) = z\phi_K(|z|) + \mathfrak{g}(z).$$

There is a constant C, independent of K, such that

$$\mathrm{Lip}(f_K) \le C(1+K)^{1+\gamma}, \tag{4.1}$$

$$|f(z) - f_K(z)| \le C(1+K)^{\gamma} e^{-K}(1+|z|^2), \tag{4.2}$$

$$|f(z)| + |f_K(z)| \le C(1+|z|^{3/4} + |z|^{5/4}). \tag{4.3}$$

Moreover, with $\Gamma_K = L_g + D(1+K)^{\gamma}$,

$$|\mathrm{Im}((f_K(z) - f_K(w))\overline{z-w})| \le \Gamma_K |z-w|^2. \tag{4.4}$$

Let $Pu = f_K(u) + H$ and $Pv = f_K(v) + J$ be Dirichlet mild solutions with integrable L^2 right-hand sides. Then

$$\|u(t) - v(t)\|_2 \le e^{\Gamma_K t}\left(\|u(0) - v(0)\|_2 + \int_0^t \|H(r) - J(r)\|_2 \, \mathrm{d}r\right). \tag{4.5}$$

Every mild solution of $Pu = f_K(u) + H$, and every mild solution with f in place of f_K and integrable nonlinearity, satisfies

$$\|u(t)\|_2 \le e^{L_g t}\left(\|u(0)\|_2 + t\ell^{1/2}|\mathfrak{g}(0)| + \int_0^t \|H(r)\|_2 \, \mathrm{d}r\right). \tag{4.6}$$

If $\gamma = 0$, (4.4) and (4.5) also hold for f with the constant $L_g + D$.

Proof. Let $\Phi(q) = \phi(e^q)$. By local absolute continuity, $|\Phi'(q)| \le D(1+|q|)^{\gamma}$ almost everywhere, and hence

$$|\phi(r)| \le |\phi(1)| + CD(1 + |\log r|)^{1+\gamma}.$$

The truncation satisfies $r|\phi_K'(r)| \le D(1+K)^{\gamma}$ almost everywhere. The associated radial map is linear near zero and locally Lipschitz elsewhere. The derivative bound $|\phi_K(r)| + r|\phi_K'(r)| \le C(1+K)^{1+\gamma}$ proves (4.1).

For the lower tail, write $r = e^{-K-b}$, $b \ge 0$. Then

$$r|\phi(r) - \phi(e^{-K})| \le De^{-K-b}\int_K^{K+b} (1+q)^{\gamma} \, \mathrm{d}q \le CD(1+K)^{\gamma} e^{-K},$$

because $b(1+b)^{\gamma}e^{-b}$ is bounded. For the upper tail, write $r = e^{K+b}$ and apply the same bound after dividing by $e^{-K}r^2$. This yields (4.2), since the perturbation cancels exactly. For the growth estimate, every fixed power of $1+|\log r|$ is dominated by $r^{-1/4}+r^{1/4}$, and the truncation reduces the absolute logarithmic amplitude.

For a real radial coefficient φ with $r|\varphi'(r)| \le B$, write $r = |z| \ge s = |w| > 0$. Then $|\varphi(r)-\varphi(s)| \le B(r-s)/s$ and $|\operatorname{Im}(z\bar{w})| \le s|z-w|$. The identity

$$\operatorname{Im}((z\varphi(r) - w\varphi(s))\overline{z-w}) = (\varphi(s)-\varphi(r))\operatorname{Im}(z\bar{w})$$

gives the one-sided bound. The case of a zero amplitude follows directly. Applying this bound to ϕ_K and adding the Lipschitz bound for $\mathfrak{g}$, we obtain the stated estimate. The interaction variables $U(-t)u(t)$ are absolutely continuous in L^2 when the right-hand side is integrable. Applying the energy identity to the difference, regularizing the norm at zero, and then using Gronwall's inequality, we obtain (4.5).

The real radial term has zero contribution to the mass identity, while $|\mathfrak{g}(z)| \le L_g|z| + |\mathfrak{g}(0)|$. This yields (4.6). When $\gamma = 0$, the original coefficient satisfies the required uniform derivative bound, so the final assertion follows. □

The following periodic restriction estimate is a basic case of the Fourier estimates in [5]. We include a proof based on the resonance relations.

Lemma 4.2. *For every fixed $T_* > 0$, there exists $B = B(\ell, T_*)$ such that for $0 < \tau \le T_*$, $v \in L^2(0,\ell)$, and $F \in L^1(0,\tau;L^2(0,\ell))$,*

$$\left\| U(t)v - i\int_0^t U(t-s)F(s)\,\mathrm{d}s \right\|_{L^4((0,\tau)\times(0,\ell))} \le B\left(\|v\|_2 + \|F\|_{L^1(0,\tau;L^2)}\right). \tag{4.7}$$

Proof. We extend v oddly to the circle of length 2ℓ. The periodic Schrödinger flow has time period $P_* = 2\ell^2/\pi$. For a Fourier polynomial, integrating the fourth power of the absolute value of its evolution over one spatial and one temporal period leaves only the resonant relations

$$n_1 + n_2 = n_3 + n_4, \qquad n_1^2 + n_2^2 = n_3^2 + n_4^2.$$

These imply $\{n_1, n_2\} = \{n_3, n_4\}$. In fact, for $V(x) = \sum_n c_n e^{in\pi x/\ell}$ and its periodic flow U_{per}, the exact identity is

$$\int_0^{P_*}\int_{-\ell}^{\ell} |U_{\mathrm{per}}(t)V|^4\,\mathrm{d}x\,\mathrm{d}t = 2\ell P_*\left[2\left(\sum_n |c_n|^2\right)^2 - \sum_n |c_n|^4\right].$$

Parseval's identity and a density argument give the free estimate over one period. The odd extension has twice the squared L^2 norm of v, and the fourth power of its evolution is even in space. Covering $[0,T_*]$ by finitely many periods gives a constant independent of $\tau \le T_*$. Restricting the flow to $(0,\ell)$ yields the Dirichlet estimate. For each fixed s, the same bound applies to $\mathbf{1}_{\{s<t<\tau\}}U(t-s)F(s)$. An application of Minkowski's inequality then gives the estimate for the Duhamel term. □

Lemma 4.3. *Let $T > 0$, $v \in L^2(0,\ell)$ and $H \in L^1(0,T;L^2(0,\ell))$. Then there exists a unique Dirichlet mild solution of $Pu = f(u) + H$, $u(0) = v$, in the class*

$$u \in C([0,T];L^2) \cap L^4((0,T)\times(0,\ell)).$$

Its nonlinearity belongs to L^2 on the cylinder, and (4.6) holds. For $0 < T \le 1$, both this solution and all its truncation approximants satisfy

$$\|u\|_{L^4} \le C(1 + M^{3/2}), \qquad M = e^{L_g T}(\|v\|_2 + T\ell^{1/2}|\mathfrak{g}(0)| + \|H\|_{L^1L^2}), \tag{4.8}$$

where C depends only on the fixed interval and nonlinearity parameters. For $\gamma = 0$, uniqueness also holds among $C([0,T];L^2)$ mild solutions with nonlinearity in L^1L^2.

Proof. Let u_K solve the globally Lipschitz equation with f_K. For each K, its nonlinearity belongs to $L^1(0,T;L^2)$. Lemma 4.2 therefore gives $u_K \in L^4$. The mass bound is uniform in K. Set $Z_K = \|u_K\|_{L^4}$. Combining Lemma 4.2 with the growth bound and interpolation, we obtain, with constants independent of K,

$$Z_K \le B[M + CT\ell^{1/2} + CT\ell^{1/8}M^{3/4} + CT^{7/8}M^{3/4}Z_K^{1/2}].$$

Indeed, the inequality $\||u_K(t)|^{5/4}\|_2 \le M^{3/4}\|u_K(t)\|_4^{1/2}$, followed by Hölder's inequality in time, yields the last term. Young's inequality then gives a uniform finite L^4 bound for fixed T, as well as (4.8) for $T \le 1$.

To compare two approximants with $J \ge K$, we rewrite both equations with f_K and treat $f_J(u_J) - f_K(u_J)$ as an additional source in the equation for u_J. The tail estimate gives

$$|f_J(z) - f_K(z)| \le C(1+K)^\gamma e^{-K}(1 + |z|^2), \qquad J \ge K,$$

since $(1+K)^\gamma e^{-K}$ is decreasing for $K \ge 1$. Equations (4.2) and (4.5) imply

$$\|u_K - u_J\|_{CL^2} \le C(1+K)^\gamma e^{-K+T\Gamma_K}(T\ell^{1/2} + \sqrt{T}\,\|u_J\|_{L^4}^2).$$

The factor on the right tends to zero because $\gamma < 1$, so the approximants converge in $C([0,T];L^2)$ to u. Passing to a subsequence gives almost everywhere convergence, and Fatou's lemma yields $u \in L^4$. To identify the limiting nonlinearity, consider first a point where $u \neq 0$. The approximants eventually lie in a compact annulus on which the radial truncation is inactive. At a point where $u = 0$, write $f_K(u_K) = F_K(u_K) + \mathfrak{g}(u_K)$, where $F_K(z) = z\phi_K(|z|)$. The bound $|F_K(z)| \le C(|z|^{3/4} + |z|^{5/4})$, without a constant term, gives $F_K(u_K) \to 0$, while continuity gives $\mathfrak{g}(u_K) \to \mathfrak{g}(0) = f(0)$. Thus $f_K(u_K) \to f(u)$ almost everywhere. For every measurable subset $E \subset (0,T)\times(0,\ell)$,

$$\int_E |f_K(u_K)|^2 \,\mathrm{d}x\,\mathrm{d}t \le C(|E| + |E|^{5/8} + |E|^{3/8}),$$

by the uniform L^4 bound. Fatou's lemma and Vitali's theorem therefore give $f(u) \in L^2$ and $f_K(u_K) \to f(u)$ strongly in L^2. The Duhamel integrals converge in CL^2. Passing to the limit in the mild formulation proves existence.

The same comparison gives uniqueness. Let u, v belong to $CL^2 \cap L^4$ and solve the equation with the same initial state and source. By (4.3), their nonlinearities belong to $L^2(Q)$. Rewriting both equations with f_K, we obtain

$$\|u - v\|_{CL^2} \le C(1+K)^\gamma e^{-K+T\Gamma_K}(T\ell^{1/2} + \sqrt{T}(\|u\|_{L^4}^2 + \|v\|_{L^4}^2)).$$

Letting $K \to \infty$, we obtain uniqueness without requiring a stability constant uniform in K. For $\gamma = 0$, the last assertion follows from the one-sided bound for the untruncated equation in Lemma 4.1. □

4.2 Subcritical control near a reference trajectory. We now apply the forced well-posedness theory to a truncated control problem. The following lemma estimates the control cost and the terminal error relative to the reference trajectory.

Lemma 4.4. *Let $u \in C([0,T];L^2) \cap L^4(Q)$ solve $Pu = f(u)$. There exist $C > 0$ and $B \geq 1$ such that, for every interval $I = [t_*, t_* + \tau] \subset [0,T]$ with $0 < \tau \leq 1$, every $K \geq 1$ and every $v \in L^2$, one can choose a control h_K with the following properties. Set*

$$q = \frac{2(1+\gamma)}{3} < 1, \qquad E_K(\tau) = C(\tau^{-1} + K^q), \qquad M = \|v\|_2.$$

The solution of the shifted truncated problem

$$Pz_K = f_K(u + z_K) - f_K(u) + \mathbf{1}_\omega h_K, \qquad z_K(t_*) = v,$$

satisfies $z_K(t_* + \tau) = 0$ and

$$\|h_K\|_{L^2(I\times\omega)} \leq e^{E_K(\tau)} M, \tag{4.9}$$

$$\|z_K\|_{C(I;L^2)} + \|z_K\|_{L^4(I\times\Omega)} \leq e^{E_K(\tau)} M. \tag{4.10}$$

For the solution $Py_K = f(y_K) + \mathbf{1}_\omega h_K$ of the untruncated equation with $y_K(t_*) = u(t_*) + v$,

$$\|y_K(t_* + \tau) - u(t_* + \tau)\|_2 \leq e^{-K + E_K(\tau)}(B + M^3). \tag{4.11}$$

The constant C depends only on the fixed interval, control region and nonlinearity parameters. One may choose

$$B = C(1 + \|u\|^3_{C([0,T];L^2)} + \|u\|^2_{L^4(Q)}), \tag{4.12}$$

after increasing C to ensure $B \geq 1$.

Proof. The function f_K is globally Lipschitz with constant at most $C(1+K)^{1+\gamma}$. Applying Lemma 3.9 to $f_K(u+z) - f_K(u)$ on I after a time translation, we obtain a null control and the CL^2 estimate with exponent $C(\tau^{-1} + K^q)$. To estimate the space–time norm, we use the interval L^4 bound and obtain

$$\|z_K\|_{L^4} \leq B_\ell(M + \tau \operatorname{Lip}(f_K)\|z_K\|_{CL^2} + \sqrt{\tau}\|h_K\|_2).$$

Absorbing the polynomial factor into the exponential, we obtain (4.9)–(4.10) after enlarging C; this constant remains independent of the reference.

Lemma 4.3 gives the solution of the untruncated equation with the same control. Combining its mass and L^4 estimates, we obtain

$$\|y_K\|^2_{L^4(I\times\Omega)} \leq C\left[1 + (\|u(t_*)\|_2 + M + \sqrt{\tau}\|h_K\|_2)^3\right].$$

Here the constants absorb the fixed quantities L_g and $\ell^{1/2}|\mathfrak{g}(0)|$, since $\tau \leq 1$. It remains to compare y_K with $u + z_K$. Writing both equations with f_K leaves a difference between their source terms given by

$$[f(y_K) - f_K(y_K)] - [f(u) - f_K(u)].$$

Their initial states agree. It therefore follows from (4.5) and the tail bound that

$$\|y_K - u - z_K\|_{C(I;L^2)} \leq C(1+K)^\gamma e^{-K + \tau\Gamma_K}\left[\tau\ell^{1/2} + \sqrt{\tau}(\|y_K\|^2_{L^4(I\times\Omega)} + \|u\|^2_{L^4(I\times\Omega)})\right].$$

Since $\gamma < q$ and $\tau \leq 1$, we absorb the factors $(1+K)^\gamma$ and $e^{\tau\Gamma_K}$, together with the preceding state bound, into $e^{C(\tau^{-1}+K^q)}$. Evaluating at the terminal time, using $z_K(t_* + \tau) = 0$ and enlarging C, we obtain (4.11) with (4.12). □

Proof of Theorem 1.8(i). The class of nonlinearities is invariant under $f^\sharp(z) = \overline{f(\bar z)}$: the real radial part is unchanged and $\mathfrak{g}$ is replaced by the globally Lipschitz function $\overline{\mathfrak{g}(\bar z)}$. We solve the uncontrolled $f^\sharp$ equation forward from $\overline{y_1}$ and conjugate its time reversal. This gives an uncontrolled reference $u \in C([0,T];L^2)\cap L^4(Q)$ for f, with $u(T)=y_1$.

Set $S=\min\{T,1\}$. We construct a control joining y_0 to $u(S)$ by time S. To this end, put

$$a=\frac{6S}{\pi^2},\qquad \tau_j=\frac{a}{(j+1)^2},\qquad t_0=0,\quad t_{j+1}=t_j+\tau_j,\qquad K_j=A2^j.$$

Then $t_j\uparrow S$. Let C,B be as in Lemma 4.4, and put $M_0=\|y_0-u(0)\|_2$. Choose $A\ge 1$ so that

$$C(\tau_j^{-1}+K_j^q)\le K_j/8\quad (j\ge 0),\qquad \log(B+\max\{1,M_0^3\})\le A/8. \tag{4.13}$$

Such a choice is possible because $q<1$. More explicitly, $\sup_{j\ge0}(j+1)^2/2^j=9/4$, so it suffices that

$$A\ge \max\left\{1,\frac{36C}{a},(16C)^{1/(1-q)},8\log(B+\max\{1,M_0^3\})\right\}.$$

We apply the preceding reference-trajectory lemma on each successive interval and use the resulting control in the true equation. Set $M_j=\|y(t_j)-u(t_j)\|_2$ and $H_j=\|h_j\|_{L^2((t_j,t_{j+1})\times\omega)}$. Then

$$H_j\le e^{K_j/8}M_j,\qquad M_{j+1}\le e^{-7K_j/8}(B+M_j^3).$$

The parameter choice gives $M_1\le e^{-3K_0/4}$. Induction yields

$$M_{j+1}\le e^{-K_j/2},\qquad H_j\le e^{K_j/8-K_{j-1}/2}=e^{-K_j/8}\quad (j\ge1).$$

Indeed, once $M_j<1$, $\log(B+M_j^3)\le\log(B+1)\le A/8\le K_j/8$. Consequently

$$\sum_{j\ge0}H_j^2\le e^{A/4}M_0^2+\sum_{j\ge1}e^{-A2^j/4}<\infty.$$

The concatenated control therefore belongs to L^2. We extend it by zero on (S,T) and apply Lemma 4.3 on the full interval. Uniqueness identifies this global solution with each successive local solution. Its continuity and $y(t_j)-u(t_j)\to0$ imply $y(S)=u(S)$. Uniqueness on $[S,T]$ then gives $y(T)=u(T)=y_1$. The global well-posedness lemma also gives the asserted L^4 and nonlinear L^2 regularity. This continuity argument identifies the accumulated endpoint without requiring a mass estimate for the difference $y-u$.

We next prove the quantitative assertion. Let $0<T\le1$ and $d=\|y_0\|_2+\|y_1\|_2$. The backward reference satisfies

$$\|u\|_{CL^2}\le C(1+d),\qquad \|u\|_{L^4}^2\le C(1+d^3).$$

Thus $B\le C(1+d^3)$ and $M_0\le C(1+d)$. Put $W=B+\max\{1,M_0^3\}$. All requirements in (4.13) hold for

$$A=C_0/T+C_1+8\log W$$

with sufficiently large fixed C_0,C_1. In particular,

$$A\le C(T^{-1}+\log(2+d)).$$

We retain the original control estimate on the first interval, rather than replacing its exponent by $A/8$. Since $\tau_0=6T/\pi^2$, we obtain

$$\begin{aligned}H_0&\le\exp\bigl(C(\tau_0^{-1}+A^q)\bigr)\,M_0\\&\le Ce^{C/T}(1+d)\exp\bigl(C[\log(2+d)]^q\bigr).\end{aligned}$$

Here we used $(a+b)^q \le a^q + b^q$ for $a, b \ge 0$, and $T^{-q} \le T^{-1}$. The sum of the squared norms of the remaining controls is bounded independently of d and T. The mass estimate (4.6) gives the same bound for the supremum norm of the state. Thus (1.8) holds for $0 < T \le 1$.

For $T > 1$, let $J = \lceil T \rceil$ and $\tau = T/J \in (1/2, 1]$. We apply the preceding estimate on the J successive intervals, joining y_0 to zero on the first, zero to y_1 on the last, and zero to zero on each intervening interval. Since the interval lengths satisfy $\tau^{-1} < 2$, the constants are uniform over these controls. The squared control norms add, and the state norm is the maximum of the segment norms. Hence

$$\begin{aligned} \|h\|^2_{L^2(Q_\omega)} &\le C(1+d)^2 \exp(2C[\log(2+d)]^q) + CJ, \\ \|y\|_{C([0,T];L^2)} &\le C(1+d) \exp(C[\log(2+d)]^q)\,. \end{aligned}$$

Since $J \le T + 1 \le 2T$, these bounds give (1.8) for $T > 1$. The states match at every junction, so their finite concatenation is a mild solution and retains the stated L^4 and nonlinear L^2 regularity. When $f(0) \neq 0$, the intermediate trajectories and controls need not vanish; their contribution gives the $\sqrt{T}$ term.

Finally, for every $\varepsilon > 0$, the inequality $q < 1$ gives

$$\sup_{r \ge 0} (Cr^q - \varepsilon r) < \infty.$$

Applying this with $r = \log(2+d)$ and using $2 + d \le 2(1+d)$, we obtain (1.9).

□

Proof of Theorem 1.8(ii). For $0 < \sigma \le 1$, Theorem 1.8(i) gives a uniform null-control cost $C_\lambda e^{C_\lambda/\sigma}$ on the unit ball. For $a > 0$, the transformation

$$(y, h) \longmapsto (ae^{-i\lambda \log(a^2)t} y, ae^{-i\lambda \log(a^2)t} h)$$

preserves this equation and multiplies all relevant norms by a. Applying this scaling to normalized nonzero initial states gives a null control with cost linear in the initial norm. For the zero state, we use the zero control. For arbitrary $T > 0$, set $\sigma = \min\{T/2, 1/2\}$. Join y_0 to zero on $[0, \sigma]$, and use the conjugate time reversal of a null trajectory from $\overline{y_1}$ on $[T - \sigma, T]$. On the intervening interval, take the state and control to be zero, which is possible since $f(0) = 0$. The equation has the conjugation symmetry required for this time reversal. The squared control norms add, and the state norm is the maximum over the two controlled intervals. Since $\sigma^{-1} = 2\max\{T^{-1}, 1\} \le 2(T^{-1} + 1)$, increasing C_λ gives (1.10) for every $T > 0$. □

Remark 4.5. The comparison between the control cost and the truncation error relies on $K^q = o(K)$, where $q = 2(1+\gamma)/3 < 1$. The nonuniform stability factor $e^{C\tau K^\gamma}$ also has a sublinear exponent and is included in the error estimate. The argument therefore covers logarithmic powers strictly below $3/2$, including those with a nonconservative Lipschitz perturbation. It does not apply to polynomial nonlinearities. The critical logarithmic exponent is treated below under a smallness condition; the case of arbitrary critical coefficients remains open. For $\gamma > 0$, the uniqueness class includes the L^4 condition needed for the tail comparison. Uniqueness in general $C_t L^2_x$ mild classes is not asserted.

4.3 The critical logarithmic exponent. At $\gamma = 1/2$, the exponent in the truncated control cost is linear in K. We therefore keep its leading coefficient explicit and show that smallness of this coefficient allows the truncation error to decay fast enough for the cascade. We first treat the pure critical derivative bound and then include the arbitrary lower-order term in (1.11).

Proof of Theorem 1.9. Choose $C_* \geq 1$ so that (3.32) holds with C_* in both occurrences of C, and set

$$D_* := \frac{3}{5}\left(\frac{1}{64C_*}\right)^{3/2}, \qquad \mathfrak{a}(D) := C_*\left(\frac{5D}{3}\right)^{2/3}, \qquad c_* := 24C_*\left(\frac{5}{3}\right)^{2/3}.$$

Step 1. A pure critical derivative bound. Assume first that $D_0 = 0$, and write $D = D_1$. Thus

$$r|\phi'(r)| \leq D\sqrt{1 + |\log r|}, \qquad 0 \leq D \leq D_*. \tag{4.14}$$

The truncation, mass, and growth bounds in Lemma 4.1 remain valid when $\gamma = 1/2$. The proof of Lemma 4.3 also applies here: the Cauchy and uniqueness comparisons require only $\gamma < 1$, since $(1+K)^\gamma e^{-K+T(L_g+D(1+K)^\gamma)} \to 0$. The lemma therefore gives the global forced solution and the uniform bound $\|y\|_{L^4} \leq C(1 + M^{3/2})$. We keep $L_g = \mathrm{Lip}(\mathfrak{g})$ fixed.

To compare the control cost with the truncation error, we retain the dependence of the control exponent on D. Integration of (4.14) gives

$$|\phi_K(r)| \leq |\phi(1)| + \frac{2D}{3}(1+K)^{3/2}, \qquad r|\phi_K'(r)| \leq D(1+K)^{1/2}.$$

Consequently, with $L_0 = |\phi(1)| + L_g$,

$$L_K := \mathrm{Lip}(f_K) \leq L_0 + \frac{5D}{3}(1+K)^{3/2}, \qquad L_K^{2/3} \leq L_0^{2/3} + \left(\frac{5D}{3}\right)^{2/3}(1+K).$$

Let u be an uncontrolled reference trajectory. On any interval $I = [t_*, t_* + \tau]$, $0 < \tau \leq 1$, we apply Lemma 3.9 to $f_K(u+z) - f_K(u)$, with initial state v and terminal state zero. This shifted nonlinearity has Lipschitz constant at most L_K and vanishes at zero. Writing $a = \mathfrak{a}(D)$, we obtain

$$\|h_K\|_{L^2(I\times\omega)} + \|z_K\|_{C(I;L^2)} \leq e^{c_0/\tau + aK}M, \qquad M = \|v\|_2, \tag{4.15}$$

where, for example, one may take

$$c_0 = C_* + \log C_* + C_* L_0^{2/3} + a.$$

The coefficient of K in this estimate depends only on D and the geometry. In particular, increasing c_0 does not increase a. The shifted nonlinearity belongs to $L^1(I;L^2)$, since its modulus is at most $L_K|z_K|$. Lemma 4.2 therefore gives $z_K \in L^4(I\times\Omega)$.

Let y_K solve the true equation with this control and initial state $u(t_*) + v$. Put

$$R = \|u\|_{C([0,T];L^2)}, \qquad B = 1 + R^3 + \|u\|_{L^4(Q)}^2 \geq 1.$$

The mass estimate and the uniform L^4 bound imply

$$\|y_K\|_{L^4(I\times\Omega)}^2 \leq C\left(1 + R^3 + e^{3c_0/\tau + 3aK}M^3\right).$$

We now estimate the truncation error. As in Lemma 4.4, we compare y_K and $u + z_K$ using the same nonlinearity f_K. Their initial states agree, and their source terms differ by $[f(y_K) - f_K(y_K)] - [f(u) - f_K(u)]$. Thus

$$\begin{aligned}\|y_K - u - z_K\|_{C(I;L^2)} \leq{}& C(1+K)^{1/2}e^{-K+\tau(L_g+D\sqrt{1+K})} \\ &\times \left[\tau\ell^{1/2} + \sqrt{\tau}(\|y_K\|_{L^4(I\times\Omega)}^2 + \|u\|_{L^4(I\times\Omega)}^2)\right].\end{aligned}$$

For every $\eta > 0$,

$$(1+K)^{1/2} e^{D\sqrt{1+K}} \le C_{D,\eta} e^{\eta K} \qquad (K \ge 1).$$

Since $\tau \le 1$, we absorb all factors independent of K into $e^{c_\eta/\tau}$, with $c_\eta \ge c_0$. Evaluating at the terminal time, where $z_K = 0$, gives

$$\|y_K(t_* + \tau) - u(t_* + \tau)\|_2 \le e^{c_\eta/\tau - (1-3a-\eta)K}(B + M^3). \tag{4.16}$$

Here c_η depends only on the fixed geometry, the displayed nonlinearity parameters and η; it is independent of u, K, τ, and v.

We construct the backward reference from y_1 using the conjugate nonlinearity, as in the proof of Theorem 1.8. Set $S = \min\{T, 1\}$ and

$$b = \frac{6S}{\pi^2}, \qquad \tau_j = \frac{b}{(j+1)^2}, \qquad K_j = A2^j, \qquad M_0 = \|y_0 - u(0)\|_2.$$

Set $t_0 = 0$ and $t_{j+1} = t_j + \tau_j$, so that $t_j \uparrow S$. On each interval, we choose the preceding control and write

$$M_j = \|y(t_j) - u(t_j)\|_2, \qquad H_j = \|h_j\|_{L^2((t_j, t_{j+1})\times\omega)}.$$

Fix $\eta = 1/64$. Our smallness assumption gives $a \le 1/64$. Choose

$$A = 1 + \frac{36 c_\eta}{b} + 8 \log(B + \max\{1, M_0^3\}). \tag{4.17}$$

Since $\sup_{j\ge 0}(j+1)^2/2^j = 9/4$, this ensures $c_\eta/\tau_j \le K_j/16$. Moreover, $3a + \eta \le 1/16$. Equations (4.15) and (4.16) therefore give the recursive bounds used in the subcritical cascade:

$$H_j \le e^{K_j/8} M_j, \qquad M_{j+1} \le e^{-7K_j/8}(B + M_j^3).$$

The logarithmic term in (4.17) gives $M_1 \le e^{-3K_0/4}$. Once $M_j \le 1$, $\log(B + M_j^3) \le A/8 \le K_j/8$. Induction therefore yields

$$M_{j+1} \le e^{-K_j/2}, \qquad H_j \le e^{-K_j/8} \quad (j \ge 1).$$

Thus $\sum_j H_j^2 < \infty$. We extend the concatenated control by zero on (S, T) and apply Lemma 4.3 on $[0, T]$. By uniqueness, this global forced solution agrees with each local trajectory. Its continuity gives $y(S) = u(S)$, and uniqueness on $[S, T]$ then yields $y(T) = y_1$. The global well-posedness lemma also gives the asserted L^4 and nonlinear L^2 regularity.

To estimate the cost, suppose $T \le 1$ and set $d = \|y_0\|_2 + \|y_1\|_2$. The reference bounds imply

$$B \le C(1 + d^3), \qquad M_0 \le C(1 + d), \qquad W := B + \max\{1, M_0^3\} \le C(1 + d)^3.$$

We retain the coefficient a in the first control estimate. Since $\tau_0 = b = 6T/\pi^2$, we obtain

$$\begin{aligned} H_0 &\le e^{c_0/b + aA} M_0 \le C e^{C/T} W^{8a} M_0 \\ &\le C e^{C/T}(1 + d)^{1+24a}. \end{aligned}$$

The sum of the squared norms of the remaining controls is uniformly bounded. The mass estimate gives the same bound for the supremum norm of the state. We therefore obtain

$$\|h\|_{L^2(Q_\omega)} + \|y\|_{C([0,T];L^2)} \le C e^{C/T}(1 + d)^{1+24\mathfrak{a}(D)}. \tag{4.18}$$

Step 2. An arbitrary lower-order term. Set $a = \mathfrak{a}(D_1)$ and $L_0 = |\phi(1)| + L_g$. We use the backward reference construction from Step 1, with $M_0 = \|y_0 - u(0)\|_2$ and $B = 1 + \|u\|_{CL^2}^3 + \|u\|_{L^4(Q)}^2$. The forced well-posedness and tail estimates hold with the unrestricted constant $D_0 + D_1$ and exponent $\gamma = 1/2$. To estimate the control cost, we keep the separate contributions in the Lipschitz bound:

$$L_K \le L_0 + D_0(1+K) + \frac{5D_1}{3}(1+K)^{3/2}.$$

Subadditivity of the power $2/3$ gives, with $b = C_* D_0^{2/3}$,

$$\|h_K\|_2 + \|z_K\|_{CL^2} \le e^{c_0/\tau + aK + bK^{2/3}} M.$$

Here c_0 absorbs the constant terms from $(1+K)^{2/3} \le 1 + K^{2/3}$; the coefficients a and b remain unchanged. The comparison with the true equation uses the same truncation as in Step 1 and gives, for every $\eta > 0$,

$$M_{\text{next}} \le e^{c_\eta/\tau - (1-3a-\eta)K + 3bK^{2/3}}(B + M^3).$$

Fix $\eta = 1/64$. Since $3bK^{2/3} \le K/64 + C_b$, the constant C_b can be absorbed into $\bar{c}/\tau$, with $\bar{c} \ge c_\eta, c_0$ depending only on the fixed parameters. The same choice of $\bar{c}$ also gives

$$\|h_K\|_2 \le e^{\bar{c}/\tau + (a+1/192)K} M, \qquad M_{\text{next}} \le e^{\bar{c}/\tau - (1-3a-1/32)K}(B + M^3).$$

Set $s = 6\min\{T, 1\}/\pi^2$, $\tau_j = s/(j+1)^2$, and $K_j = A2^j$. Choose the cascade parameter

$$A = 1 + \frac{72\bar{c}}{s} + 8\log(B + \max\{1, M_0^3\}).$$

Then $\bar{c}/\tau_j \le K_j/32$, while $a \le 1/64$. The control exponent is at most $(1/32 + 1/64 + 1/192)K_j < K_j/8$, and the loss in the terminal-error exponent is at most $(1/32 + 3/64 + 1/32)K_j = 7K_j/64 < K_j/8$. These bounds allow us to use the recursive estimates and the infinite concatenation from Step 1.

For $T \le 1$, we have $A \le C(T^{-1} + \log(2+d))$. To estimate the first control, we retain the unabsorbed $bA^{2/3}$ term and obtain

$$H_0 \le e^{c_0/\tau_0 + aA + bA^{2/3}} M_0 \le C e^{C/T}(1+d)^{1+24a} e^{C[\log(2+d)]^{2/3}}.$$

Estimating the remaining controls and the state as before, we obtain

$$\|h\|_{L^2(Q_\omega)} + \|y\|_{C([0,T];L^2)} \le C e^{C/T}(1+d)^{1+24a} \exp\big(C[\log(2+d)]^{2/3}\big). \tag{4.19}$$

Since $24a = c_* D_1^{2/3}$, we absorb the final factor into $C_\varepsilon(1+d)^\varepsilon$ to obtain (1.12) for $0 < T \le 1$. For $T > 1$, we use the finite concatenation from the proof of Theorem 1.8(i), with $J = \lceil T \rceil$ and $\tau = T/J \in (1/2, 1]$. The first and last intervals contribute at most $C_\varepsilon(1+d)^{2(1 + c_* D_1^{2/3} + \varepsilon)}$ to the squared control norm, and the remaining intervals contribute at most $C_\varepsilon J$. The state norm is bounded by the maximum of the segment norms. Taking square roots and using $J \le 2T$, we obtain (1.12) for all $T > 0$, with the same smallness threshold D_* and exponent $c_* D_1^{2/3}$. This completes the proof. □

A Verification of the planar phase condition

In this appendix, we shall verify Assumption 1.2 for the control region defined in (1.6). The uncontrolled compact set is $K_\omega = K_0 \cup K_1$, where

$$K_0 = \{x \in \overline{\Omega} : d(x) \geq 2\delta\}, \qquad K_1 = \{x \in \overline{\Omega} : d(x) \leq \delta,\ \pi(x) \notin \Gamma_c\}.$$

The two sets are separated by a positive distance.

Near K_0, take

$$\psi_0(x) = |x - z|^2, \qquad z \in \mathbb{R}^2 \setminus \overline{\Omega}.$$

Then $D^2\psi_0 = 2I$, and $\nabla\psi_0$ is bounded away from zero near K_0.

For the phase near K_1, cut the boundary at a point strictly inside Γ_c. The resulting arclength coordinate s is single-valued on a neighborhood of $\partial\Omega \setminus \Gamma_c$. Let $\gamma(s)$ be the boundary parametrization, $\tau = \gamma'$, and ν the outward unit normal. Define the signed curvature by $\nu' = k\tau$. In the normal coordinates

$$X(s, r) = \gamma(s) - r\nu(s),$$

where $r > 0$ inside Ω, set

$$\psi_1(X(s, r)) = \exp(as + br^2), \qquad a > 0, \quad 2b > \|k\|^2_{L^\infty(\partial\Omega)}.$$

This formula also defines ψ_1 smoothly for small negative r. At $r = 0$,

$$\partial_\nu\psi_1 = 0, \qquad \nabla\psi_1 = ae^{as}\tau, \qquad D^2\psi_1 = e^{as}\begin{pmatrix} a^2 & -ak \\ -ak & 2b \end{pmatrix} \quad \text{in the frame } (\tau, \nu).$$

The first diagonal entry is positive, and the determinant is

$$a^2e^{2as}(2b - k^2) > 0.$$

By compactness of $\partial\Omega \setminus \Gamma_c$, the Hessian is uniformly positive definite and the gradient is bounded away from zero on this boundary arc. With a, b fixed, continuity preserves these properties near K_1 when $\delta > 0$ is sufficiently small.

We combine the two local phases using disjoint open neighborhoods $U_j \subset \mathbb{R}^2$ of K_j, $j = 0, 1$, with $U_0 \Subset \Omega$ and U_1 contained in the normal coordinate chart. Take $\chi_j \in C_c^\infty(U_j)$ equal to one near K_j, and define

$$\psi = \chi_0\psi_0 + \chi_1\psi_1,$$

where each product is extended by zero outside U_j. Then ψ is real-valued and smooth on $\mathbb{R}^2$, and $\psi = \psi_j$ near K_j.

By compactness, there exist $c_0, c_1 > 0$ and a relatively open neighborhood $\mathcal{O}$ of K_ω in $\overline{\Omega}$ such that

$$|\nabla\psi| \geq c_0, \qquad D^2\psi \geq c_1 I \quad \text{on } \mathcal{O},$$
$$\partial_\nu\psi = 0 \quad \text{on } \mathcal{O} \cap \partial\Omega.$$

The boundary part of $\mathcal{O}$ lies where $\psi = \psi_1$, and hence the normal derivative vanishes. This verifies Assumption 1.2.

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Conflict of interest

The authors declare no conflict of interest.

Data availability

No data were generated or used in this study.

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